

A Summary for Quant interview Questions of Option Pricing

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1 Basic Derivatives

1.1 Price direction of options

How do vanilla European/American option prices change when S, K, τ, σ, r , or D changes?

solution:

Variable	European call	European put	American call	American Put
Stock price $\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$
Strike price $\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$
Time to maturity $\uparrow$	?	?	$\uparrow$	$\uparrow$
Volatility $\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
Risk-free rate $\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$
Dividends $\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$

1.2 Put-call parity

Can you write down the put-call parity for European options on non-dividend paying stocks and prove it?

solution:

Put-call parity: $c + Ke^{-rT} \equiv p + S$, where the European call option and the European put option have the same underlying security, the same maturity T and the same strike price K .

To simplify the problem, we will focus on underlying assets that do not pay dividends. To prove this relationship, we can consider an investment portfolio consisting of the following assets:

- Buy a put option;
- Buy one unit of the underlying asset;
- Sell a call option.

The value of the investment portfolio at time t is:

$$V_{\text{portfolio}}(t) = P(t) + S(t) - C(t).$$

The option values at maturity are:

$$C(T) = \max(S(T) - K, 0), \quad P(T) = \max(K - S(T), 0).$$

Therefore, regardless of whether $S(T) < K$ or $S(T) \geq K$, the value of the portfolio at time T will always be equal to K .

If $S(T) < K$, then $P(T) = K - S(T)$ and $C(T) = 0$, hence:

$$V_{\text{portfolio}}(T) = P(T) + S(T) - C(T) = (K - S(T)) + S(T) = K.$$

If $S(T) \geq K$, then $P(T) = 0$ and $C(T) = S(T) - K$, hence:

$$V_{\text{portfolio}}(T) = P(T) + S(T) - C(T) = 0 + S(T) - (S(T) - K) = K.$$

Therefore, no matter how the stock price fluctuates,

$$V_{\text{portfolio}}(T) = P(T) + S(T) - C(T) = K.$$

According to the principle of no arbitrage, the value of the portfolio at time t must be equal to the present value of K at time t , i.e.,

$$V_{\text{portfolio}}(t) = Ke^{-r(T-t)}.$$

Thus:

$$P(t) + S(t) - C(t) = Ke^{-r(T-t)}.$$

Rewriting this, we get:

$$C(t) - P(t) = S(t) - Ke^{-r(T-t)}.$$

1.3 American option 1

Since American options can be exercised at any time before maturity, they are often more valuable than European options with the same characteristics. But when the stock pays no dividend, the theoretical price for an American call and European call should be the same since it is never optimal to exercise the American call. Why should you never exercise an American call on a non-dividend paying stock before maturity?

solution:

Let us use a numerical argument based on risk-neutral pricing and **Jensen's inequality**—if $f(X)$ is a convex function, then $E[f(X)] \geq f(E[X])$. **the payoff of a call option (if exercised when $S > K$) is given by $C(S) = (S - K)^+$, which is a convex function of the stock price** and has the following property.

$$C(\lambda S_1 + (1 - \lambda)S_2) \leq \lambda C(S_1) + (1 - \lambda)C(S_2), \quad 0 < \lambda < 1.$$

Let $S_1 = S$ and $S_2 = 0$, then $C(\lambda S) \leq \lambda C(S) + (1 - \lambda)C(0) = \lambda C(S)$ since $C(0) = 0$.

If the option is exercised at time t , the payoff at t is $C(S_t - K)$. If it is not exercised until maturity, the discounted expected payoff (to t) is $\tilde{E}[e^{-r(T-t)}C(S_T)]$ under risk-neutral measure. Under risk-neutral probabilities, we also have $\tilde{E}[S_T] = S_t e^{r(T-t)}$.

$$\tilde{E}[e^{-r(T-t)}C(S_T)] = e^{-r(T-t)}\tilde{E}[C(S_T)] \geq e^{-r(T-t)}C(\tilde{E}[S_T]) = e^{-r(T-t)}C(e^{r(T-t)}S_t),$$

where the inequality is from Jensen's inequality.

Let $S = e^{r(T-t)}S_t$, and $\lambda = e^{-r(T-t)}$, we have $C(\lambda S) = C(S_t) \leq e^{-r(T-t)}C(e^{r(T-t)}S_t) \leq e^{-r(T-t)}\tilde{E}[C(S_T)]$.

Since the discounted payoff $e^{-r(T-t)}\tilde{E}[C(S_T)]$ is no less than $C(S_t)$ for any $t \leq T$ under the risk-neutral measure, it is never optimal to exercise the option before expiration.

1.4 American Option 2

Under what conditions is it not optimal to exercise an American call option early? Does the conclusion also apply to American put options?

Solution:

The basic intuition is that, in the absence of dividends paid on the underlying asset, early exercise of an American call option is usually not optimal. However, early exercise of an American put option can sometimes be optimal.

For the American call option, consider a portfolio consisting of one American call option C and a zero-coupon bond K maturing at time T with a face value of K . If the stock does not pay dividends and the stock price is higher than the strike price, the value of this portfolio at expiration is:

$$(S_T - K) + K = S_T.$$

If the stock price falls below K , the value of the portfolio is simply $\max(S_T, K)$. Thus, the portfolio always yields at least as much as owning the stock, sometimes more. Regardless of what happens in the future, the value of this portfolio must always be higher than the stock price, so:

$$C_t + KB_t > S_t, \quad \forall t < T,$$

where B_t is the value of the zero-coupon bond at time t . Rewriting this gives:

$$C_t > S_t - K.$$

Assuming the interest rate is non-negative, i.e., $B_t < 1$, we have:

$$C_t > S_t - K.$$

This inequality implies that the value of the call option is always greater than its intrinsic value. In contrast, for a put option, there is no similar conclusion, and the put option's time value may decay faster, making early exercise sometimes optimal.

Holding the American call option without dividends has more value than exercising it early because the option holder retains the upside potential while limiting the downside risk. If the option is exercised early, the option holder loses the benefit of holding the call without bearing downside risk. Early exercise of a call option can only make sense if the stock pays significant dividends, as the value of receiving dividends would be lost otherwise.

In the case of the American put option, early exercise can be rational under certain conditions. The value of the put option benefits from downward price movements, and in some cases, the holder may wish to exercise before the price moves upward again, particularly when interest rates are high or when the stock price is expected to increase before expiration.

In summary, early exercise is typically not optimal for an American call option unless the stock pays significant dividends. For American put options, early exercise can sometimes be optimal when interest rates are high or the stock price is expected to rise in the future.

1.5 European Option 1

European put option on a non-dividend paying stock with strike price 80 is currently priced at 8 and a put option on the same stock with strike price 90 is priced at 9. Is there an arbitrage opportunity existing in these two options?

solution:

In the last problem, we mentioned that the payoff of a put is a convex function in stock price. **The price of a put option as a function of the strike price is a convex function as well.** Since a put option with strike 0 is worthless, we always have $P(0) + \lambda P(K) > P(\lambda K)$.

For this specific problem, we should have $8/9 \times P(90) = 8/9 \times 9 = 8 > P(80)$. Since the put option with strike price 80 is currently priced at 8, it is overpriced and we should short it. The overall arbitrage portfolio is to short 9 units of put with $K = 80$ and long 8 units of put with $K = 90$. At time 0, the initial cash flow is 0. At the maturity date, we have three possible scenarios:

1. $S_T \geq 90$, payoff = 0 (No put is exercised.)
2. $90 > S_T \geq 80$, payoff = $8 \times (90 - S_T) > 0$ (Puts with $K = 90$ are exercised.)
3. $S_T < 80$, payoff = $8 \times (90 - S_T) - 9 \times (80 - S_T) = S_T > 0$ (All puts are exercised.)

The final payoff ≥ 0 with positive probability that payoff > 0 . So it is clearly an arbitrage opportunity.

1.6 European Option 2

Prove that the time value of a European call option is maximized at the at-the-money point (i.e., when the strike price equals the underlying asset price).

Solution:

The time value of an option is the difference between the total option price $C(S)$ and the intrinsic value $\max(S - K, 0)$. In other words, the time value is the total option value minus its intrinsic value, given by the formula:

$$C(S) - \max(S - K, 0).$$

We need to prove that the time value of the call option is maximized when $S = K$. To do this, we define the function:

$$f(S) = C(S) - \max(S - K, 0).$$

We will show that $f(S)$ achieves a maximum at $S = K$. Therefore, we need to prove that the derivative of $f(S)$ with respect to S equals zero at $S = K$. First, let's analyze the function $f(S)$ based on the value of S relative to K :

$$f(S) = \begin{cases} C(S), & \text{if } S \leq K, \\ C(S) - (S - K), & \text{if } S > K. \end{cases}$$

For $S > K$, the function $f(S)$ is the value of the call option minus the intrinsic value, so we differentiate for both cases:

$$f'(S) = \begin{cases} \Delta(S) < 1, & \text{if } S \leq K, \\ \Delta(S) - 1 < 0, & \text{if } S > K. \end{cases}$$

Since the delta of a call option always begins less than 1, and decreases after $S = K$, $f(S)$ decreases as S increases beyond K .

For the call option using Black-Scholes, we know that:

$$\Delta_{CBS} = e^{-rT} N(d_1),$$

where $N(d_1) < 1$. Therefore, $f(S)$ is maximized when $S = K$, which is the point where the option is at-the-money.

To further verify this, consider the function $g(S) = P(S) - \max(K - S, 0)$ for a put option. If $S = K$, we get:

$$g(S) = P(S).$$

If $S < K$, then:

$$g(S) = P(S) + (K - S),$$

and if $S > K$, we have:

$$g(S) = P(S).$$

For $S \leq K$, $g(S)$ is increasing since the delta of the put option starts at -1 (when volatility is high) and increases to 0 as S increases. Similarly, for call options, $f(S)$ decreases as S rises above K , meaning that the time value is maximized when $S = K$.

Thus, we have shown that the time value of a European call option is maximized when $S = K$.

1.7 Risk-Neutral Pricing

What is Risk-Neutral Pricing?

Answer:

Risk-neutral pricing, or valuation, refers to valuing derivative securities as discounted expected values of their payoffs at maturity, under the assumption that the underlying asset has lognormal distribution with a drift equal to the risk-free rate.

More precisely, if the price of the underlying asset has a lognormal distribution with volatility σ and pays dividends continuously at the rate q , then the value of a derivative security on this asset with payoff $V(T)$ at maturity T given by risk-neutral valuation is

$$V(0) = e^{-rT} \mathbb{E}_{RN}[V(S(T))],$$

where r is the risk-free rate assumed to be constant, and the expected value is computed with respect to the lognormal random variable $S(T)$ given by

$$S(T) = S(0)e^{(r-q-\frac{\sigma^2}{2})T + \sigma\sqrt{T}Z}.$$

Risk-neutral valuation can be used for derivative securities which can be perfectly hedged dynamically using cash and the underlying asset. Plain vanilla European options, as well as European options with other payoffs at maturity (such as asset-or-nothing and cash-or-nothing options) can be priced using risk-neutrality. Risk-neutral valuation cannot be used for path dependent options such as American options, barrier options, and Asian options.

1.8 Geometric Brown Motion

Assume a non-dividend paying stock follows a geometric Brownian motion. What is the value of a contract that at maturity T pays the inverse of the stock price observed at the maturity?

solution:

Under the risk-neutral measure, $dS = rSdt + \sigma SdW(t)$. Applying Itô's Lemma to $V = \frac{1}{S}$, we have:

$$dV = \left(\frac{\partial V}{\partial S} rS + \frac{\partial V}{\partial t} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S^2 \right) dt + \frac{\partial V}{\partial S} \sigma S dW(t)$$

Since $V = \frac{1}{S}$, the partial derivatives are:

$$\frac{\partial V}{\partial S} = -\frac{1}{S^2}, \quad \frac{\partial^2 V}{\partial S^2} = \frac{2}{S^3}.$$

Thus, substituting these values into the equation, we get:

$$\begin{aligned} dV &= \left(\frac{1}{S^2} rS + 0 + \frac{1}{2} \frac{2}{S^3} \sigma^2 S^2 \right) dt - \frac{1}{S^2} \sigma S dW(t) \\ &= (-r + \sigma^2) V dt - \sigma V dW(t). \end{aligned}$$

So V follows a geometric Brownian motion as well, and we can apply Itô's Lemma to $\ln V$:

$$\begin{aligned} d(\ln V) &= \left(\frac{V}{V} (-r + \sigma^2) + 0 - \frac{1}{2} \frac{V^2}{V^2} \sigma^2 \right) dt + \frac{V}{V} \sigma dW(t) \\ &= \left(-r + \frac{1}{2} \sigma^2 \right) dt - \sigma dW(t). \end{aligned}$$

Hence, $\ln(V_T) \sim \mathcal{N}(\ln(V_t) + (-r + \frac{1}{2}\sigma^2)\tau, \sigma^2\tau)$ and

$$E[V_T] = E[e^{\ln V_T}] = \frac{1}{S_t} e^{-r\tau + \sigma^2\tau}.$$

Discounting the payoff by $e^{-r\tau}$, we have:

$$V = e^{-r\tau} E[V_T] = \frac{1}{S_t} e^{-2r\tau + \sigma^2\tau}.$$

2 Binomial Tree

2.1 Binomial Tree formula

We can generalize the no-arbitrage argument just presented by considering a stock whose price is S_0 and an option on the stock (or any derivative dependent on the stock) whose current price is f . We suppose that the option lasts for time T and that during the life of the option the stock price can either move up from S_0 to a new level, S_0u , where $u > 1$, or down from S_0 to a new level, S_0d , where $d < 1$.

If the stock price moves up to S_0u , we suppose that the payoff from the option is f_u ; if the stock price moves down to S_0d , we suppose the payoff from the option is f_d .

As before, we imagine a portfolio consisting of a long position in Δ shares and a short position in one option. We calculate the value of Δ that makes the portfolio riskless. If there is an up movement in the stock price, the value of the portfolio at the end of the life of the option is

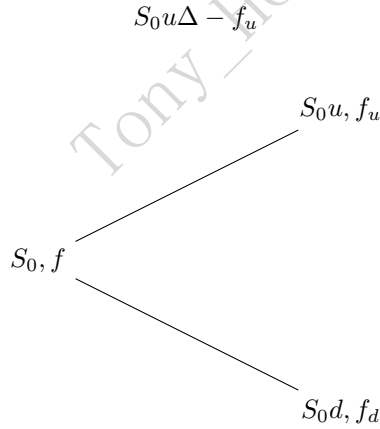


Figure 1: Stock and option prices in a general one-step tree.

If there is a down movement in the stock price, the value becomes

$$S_0d\Delta - f_d$$

The two are equal when

$$S_0u\Delta - f_u = S_0d\Delta - f_d$$

or

$$\Delta = \frac{f_u - f_d}{S_0u - S_0d} \quad (1)$$

In this case, the portfolio is riskless and, for there to be no arbitrage opportunities, it must earn the risk-free interest rate. Equation (1) shows that Δ is the ratio of the change in the option price to the change in the stock price as we move between the nodes at time T .

If we denote the risk-free interest rate by r , the present value of the portfolio is

$$(S_0u\Delta - f_u)e^{-rT}$$

The cost of setting up the portfolio is

$$S_0\Delta - f$$

It follows that

$$S_0\Delta - f = (S_0u\Delta - f_u)e^{-rT}$$

or

$$f = S_0\Delta(1 - ue^{-rT}) + f_ue^{-rT}$$

Substituting from equation (13.1) for Δ , we obtain

$$f = S_0 \left(\frac{f_u - f_d}{S_0 u - S_0 d} \right) (1 - ue^{-rT}) + f_u e^{-rT}$$

or

$$f = \frac{f_u(1 - de^{-rT}) + f_d(ue^{-rT} - 1)}{u - d}$$

or

$$f = e^{-rT} [pf_u + (1 - p)f_d]$$

where

$$p = \frac{e^{rT} - d}{u - d}$$

- e^{rT} represents the expected future price based on the risk-free rate,
- $u > 1$, the upward movement factor, is typically $u = e^{\sigma\sqrt{\Delta t}}$ where σ is the volatility of the underlying asset,
- $d < 1$, the downward movement factor, is typically $d = \frac{1}{u}$ to avoid arbitrage.

2.2 Binomial Tree Calculation

A stock is currently priced at 50 USD. In three months, the stock price will either rise to 52 USD or fall to 47 USD, with a volatility of 50%. Assume the stock does not pay dividends, and the interest rate is 0%. Compute the price of an at-the-money put option.

Solution:

We first use the risk-neutral binomial model to calculate the option price. In this case, the interest rate r is zero, so we use the following equation to compute the present value of the put option:

$$P(0) = p_{\text{up}}P_{\text{up}} + p_{\text{down}}P_{\text{down}},$$

where $P(0)$ is the present value of the option, P_{up} and P_{down} are the option values when the stock price rises and falls, respectively, and p_{up} and p_{down} are the risk-neutral probabilities of the stock price going up or down, respectively.

The risk-neutral probabilities are calculated as follows:

$$p_{\text{up}} = \frac{1 - d}{u - d}, \quad p_{\text{down}} = \frac{u - 1}{u - d},$$

where $u = \frac{52}{50} = 1.04$ and $d = \frac{47}{50} = 0.94$.

Thus, we have:

$$p_{\text{up}} = \frac{1 - 0.94}{1.04 - 0.94} = 0.6, \quad p_{\text{down}} = \frac{1.04 - 1}{1.04 - 0.94} = 0.4.$$

Now, we calculate the option payoffs. If the stock price goes up to 52 USD, the payoff of the put option will be:

$$P_{\text{up}} = \max(50 - 52, 0) = 0.$$

If the stock price falls to 47 USD, the payoff of the put option will be:

$$P_{\text{down}} = \max(50 - 47, 0) = 3.$$

Now, we can calculate the present value of the option:

$$P(0) = 0.6 \times 0 + 0.4 \times 3 = 1.2.$$

Therefore, the fair price of the put option is 1.20 USD.

3 BSM

3.1 BSM Assumption

What are the assumptions behind the Black-Scholes formula?

solution:

The original Black-Scholes formula for European calls and puts consists of the equations which require the following assumptions:

1. The stock pays no dividends.
2. The risk-free interest rate is constant and known.
3. The stock price follows a geometric Brownian motion with constant drift μ and volatility σ :

$$dS = \mu S dt + \sigma S dW(t).$$

4. There are no transaction costs or taxes; the proceeds of short selling can be fully invested.
5. All securities are perfectly divisible. (buy or sell at any amount)
6. There are no risk-free arbitrage opportunities.

3.2 BSM differential equation

Can you write down the Black-Scholes-Merton differential equation and briefly explain how to derive it?

solution:

If the evolution of the stock price is a geometric Brownian motion,

$$dS = \mu S dt + \sigma S dW(t),$$

and the derivative $V = V(S, t)$ is a function of S and t , then applying Itô's lemma yields:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW(t),$$

where $W(t)$ is a Brownian motion.

The Black-Scholes-Merton differential equation is a partial differential equation that should be satisfied by V :

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV.$$

To derive the Black-Scholes-Merton differential equation, we build a portfolio with two components: long one unit of the derivative and short $\frac{\partial V}{\partial S}$ unit of the underlying stock.

Then the portfolio has value

$$\Pi = V - \frac{\partial V}{\partial S} S,$$

and the change of Π follows the equation:

$$d\Pi = dV - \frac{\partial V}{\partial S} dS.$$

Substituting the expressions for dV and dS , we get:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \mu S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S \frac{\partial V}{\partial S} dW(t) - \frac{\partial V}{\partial S} (\mu S dt + \sigma S dW(t)).$$

Simplifying this expression:

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt.$$

It is apparent that this portfolio is risk-free since it has no diffusion term. It should have a risk-free rate of return as well, thus:

$$d\Pi = r \left(V - \frac{\partial V}{\partial S} S \right) dt.$$

Combining these results, we have:

$$\left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt = r \left(V - \frac{\partial V}{\partial S} S \right) dt.$$

Expanding and rearranging:

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV,$$

which is the Black-Scholes-Merton differential equation.

3.3 BSM math formula

Call Option Pricing Formula

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

Put Option Pricing Formula

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

Where:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma \sqrt{T}}$$

$$d_2 = d_1 - \sigma \sqrt{T}$$

Note:

- C : Call option price
- P : Put option price
- S_0 : Current stock price
- K : Strike price
- r : Risk-free interest rate
- T : Time to maturity (in years)
- σ : Volatility of the stock price
- $N(d_1), N(d_2)$: Cumulative distribution function of the standard normal distribution

The Black-Scholes-Merton model assumes that the stock price follows a geometric Brownian motion with constant volatility and interest rates, and that the markets are frictionless with no arbitrage opportunities.

3.4 Explanation of the Black-Scholes Equation

Regarding the "explanation" of this equation, its meaning is not fully self-explanatory. We start from the most explicit point, which is the form of the Black-Scholes equation for an asset that does not pay dividends:

$$\frac{\partial C}{\partial t} + rS_t \frac{\partial C}{\partial S_t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 C}{\partial S_t^2} - rC = 0$$

Where:

- t is the current time,
- S_t is the current price of the asset,
- $C(t, S_t)$ is the price of the option at time t ,
- r is the risk-free interest rate,
- σ is the volatility of the asset.

This is a partial differential equation (PDE) that describes the change in the option's price as the stock price S_t changes with time t . The equation does not directly explain the meaning of each term, but together they describe how changes in the boundary conditions of the stock price affect the option's value. These boundary conditions are crucial for determining the option's price.

The derivation of the Black-Scholes equation relies on several important assumptions. The most important assumption is that the asset follows a geometric Brownian motion, with the change in the stock price S_t following the stochastic differential equation (SDE):

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

where W_t is a Wiener process (standard Brownian motion). This assumption models the random behavior of the asset price with a drift rate of r and volatility σ .

The Black-Scholes model also assumes the existence of a risk-free portfolio that earns continuous interest at the risk-free rate r .

3.5 BSM Deduction 1

How can you derive the Black-Scholes formula for a European call on a non-dividend paying stock using risk-neutral probability measure?

solution:

Under the risk-neutral probability measure, the drift of stock price becomes the risk-free interest rate $r(t)$:

$$dS = r(t)Sdt + \sigma SdW(t).$$

Risk-neutral measure allows the option to be priced as the discounted value of its expected payoff with the risk-free interest rate:

$$V(t) = E \left[e^{-\int_t^T r(u)du} V(T) \mid S(t) \right], \quad 0 \leq t \leq T,$$

where $V(T)$ is the payoff at maturity T .

When r is constant, the formula can be further simplified as

$$V(t) = e^{-r(T-t)} E[V(T) \mid S(t)].$$

Under risk-neutral probabilities,

$$dS = rSdt + \sigma SdW(t).$$

Applying Ito's lemma, we get

$$d(\ln(S)) = \left(r - \frac{\sigma^2}{2} \right) dt + \sigma dW(t) \implies \ln S_T \sim N \left(\ln S + \left(r - \frac{\sigma^2}{2} \right) (T - t), \sigma^2 (T - t) \right).$$

So

$$S_T = S e^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\varepsilon}, \quad \text{where } \varepsilon \sim N(0, 1).$$

For a European option, we have

$$V(T) = \begin{cases} S_T - K, & \text{if } S_T > K, \\ 0, & \text{otherwise.} \end{cases}$$

So

$$S e^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\varepsilon} > K \implies \varepsilon > \frac{\ln\left(\frac{K}{S}\right) - (r - \sigma^2/2)(T-t)}{\sigma\sqrt{T-t}} = -d_2.$$

Thus,

$$E[V(T) | S] = E[\max(S_T - K, 0) | S] = \int_{d_2}^{\infty} \left(S e^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\varepsilon} - K \right) \frac{1}{\sqrt{2\pi}} e^{-\varepsilon^2/2} d\varepsilon.$$

The expected payoff is:

$$E[V(T) | S] = S e^{r(T-t)} \int_{d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(\varepsilon-\sigma\sqrt{T-t})^2/2} d\varepsilon - K \int_{d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\varepsilon^2/2} d\varepsilon.$$

Let $\tilde{\varepsilon} = \varepsilon - \sigma\sqrt{T-t}$, then $d\varepsilon = d\tilde{\varepsilon}$, $\varepsilon = -d_2 \implies \tilde{\varepsilon} = -d_2 - \sigma\sqrt{T-t} = -d_1$, and we have

$$S e^{r(T-t)} \int_{d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(\varepsilon-\sigma\sqrt{T-t})^2/2} d\varepsilon = S e^{r(T-t)} N(d_1),$$

$$K \int_{-d_2}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\varepsilon^2/2} d\varepsilon = K(1 - N(-d_2)) = K N(d_2).$$

Thus,

$$E[V(T)] = S e^{r(T-t)} N(d_1) - K N(d_2) \quad \text{and} \quad V(t) = e^{-r(T-t)} E[V(T)] = S N(d_1) - K e^{-r(T-t)} N(d_2).$$

From the derivation process, it is also obvious that $1 - N(-d_2) = N(d_2)$ is the risk-neutral probability that the call option finishes in the money.

3.6 BSM Deduction 2

How do you derive the Black-Scholes formula for a European call option on a non-dividend paying stock by solving the Black-Scholes-Merton differential equation?

solution:

You can skip this problem if you don't have background in partial differential equations (PDE). One approach to solving the problem is to convert the Black-Scholes-Merton differential equation to a heat equation and then apply the boundary conditions to the heat equation to derive the Black-Scholes formula.

Let $y = \ln S$ ($S = e^y$) and $\tilde{\tau} = T - t$, then

$$\frac{\partial V}{\partial t} = -\frac{\partial V}{\partial \tilde{\tau}}, \quad \frac{\partial V}{\partial S} = \frac{1}{S} \frac{\partial V}{\partial y}, \quad \frac{\partial^2 V}{\partial S^2} = \frac{1}{S^2} \frac{\partial V}{\partial y} + \frac{1}{S^2} \frac{\partial^2 V}{\partial y^2}.$$

The Black-Scholes-Merton differential equation

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV = 0$$

can be converted to

$$-\frac{\partial V}{\partial \tilde{\tau}} + \left(r - \frac{\sigma^2}{2} \right) \frac{\partial V}{\partial y} + \frac{1}{2} \sigma^2 \frac{\partial^2 V}{\partial y^2} - rV = 0.$$

Let $u = e^{r\tilde{\tau}}V$, the equation becomes

$$-\frac{\partial u}{\partial \tilde{\tau}} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial u}{\partial y} + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial y^2} = 0.$$

Finally, let $x = y + \left(r - \frac{\sigma^2}{2}\right) \tilde{\tau} = \ln S + \left(r - \frac{\sigma^2}{2}\right) \tilde{\tau}$ and $\tau = \tilde{\tau}$, then $\frac{\partial u}{\partial y} = \frac{\partial u}{\partial x}$ and

$$-\frac{\partial u}{\partial \tau} + \left(r - \frac{\sigma^2}{2}\right) \frac{\partial u}{\partial x} + \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2} = 0 \implies \frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2}.$$

So the original equation becomes a heat/diffusion equation

$$\frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2}.$$

For the heat equation $\frac{\partial u}{\partial \tau} = \frac{1}{2}\sigma^2 \frac{\partial^2 u}{\partial x^2}$, where $u = u(x, \tau)$ is a function of time τ and space variable x , with boundary condition $u(x, 0) = u_0(x)$, the solution is

$$u(x, \tau) = \frac{1}{\sqrt{2\pi\sigma^2\tau}} \int_{-\infty}^{\infty} u_0(\psi) \exp\left(-\frac{(x-\psi)^2}{2\sigma^2\tau}\right) d\psi.$$

For European calls, the boundary condition is $u_0(S_T) = \max(S_T - K, 0)$.

$S = \exp(x - (r - 0.5\sigma^2)\tau)$. When $x = \psi$ and $\tau = 0$, $S_T = e^\psi$.

$$\begin{aligned} u(S, \tau) = u(x, \tau) &= \frac{1}{\sqrt{2\pi\sigma^2\tau}} \int_{\ln K}^{\infty} (e^\psi - K) \exp\left(-\frac{(x-\psi)^2}{2\sigma^2\tau}\right) d\psi. \\ &= \frac{1}{\sqrt{2\pi\sigma^2\tau}} \int_{\ln K}^{\infty} (e^\psi - K) \exp\left(-\frac{(x-\psi)^2}{2\sigma^2\tau}\right) d\psi. \end{aligned}$$

Let $\varepsilon = \frac{\psi-x}{\sigma\sqrt{\tau}}$, $d\varepsilon = \frac{d\psi}{\sigma\sqrt{\tau}}$, $e^\psi = e^{x+\varepsilon\sigma\sqrt{\tau}}$, and

$$\exp\left(-\frac{(x-\psi)^2}{2\sigma^2\tau}\right) = e^{-\varepsilon^2/2}.$$

When $\psi = \ln K$, $\varepsilon = \frac{\ln(K/S) - (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}} = -d_2$.

Thus,

$$u(S, \tau) = \int_{d_2}^{\infty} \left(Se^{(r-\sigma^2/2)(T-t)+\sigma\sqrt{T-t}\varepsilon} - K\right) \frac{1}{\sqrt{2\pi}} e^{-\varepsilon^2/2} d\varepsilon.$$

Now, it's clear that the equation for $u(S, \tau)$ is exactly the same as the equation for $E[V(T) | S]$ in question B. Hence, we have

$$V(S, t) = e^{-r(T-t)}u(S, \tau) = SN(d_1) - Ke^{-rT}N(d_2)$$

as well.

3.7 BSM probability density function

The Black-Scholes formula for non-dividend paying stocks assumes that the stock follows a geometric Brownian motion. Now assume that you don't know the stochastic process followed by the stock price, but you have the European call prices for all (continuous) strike prices K . Can you determine the risk-neutral probability density function of the stock price at time T ?

Solution:

The payoff of a European call at its maturity date is $\text{Max}(S_T - K, 0)$. Therefore, under the risk-neutral measure, we have

$$c = e^{-rT} \int_K^\infty (s - K) f_{S_T}(s) ds,$$

where $f_{S_T}(s)$ is the probability density function of S_T under the risk-neutral probability measure. Taking the first and second derivatives of c with respect to K , we have

$$\begin{aligned} \frac{\partial c}{\partial K} &= e^{-rT} \frac{\partial}{\partial K} \left(\int_K^\infty (s - K) f_{S_T}(s) ds \right) \\ &= e^{-rT} \left(\int_K^\infty \frac{\partial(s - K)}{\partial K} f_{S_T}(s) ds \right) = e^{-rT} (K - K) \times 1 \\ &= e^{-rT} \int_K^\infty -f_{S_T}(s) ds \end{aligned}$$

and

$$\frac{\partial^2 c}{\partial K^2} = \frac{\partial}{\partial K} \left(\frac{\partial c}{\partial K} \right) = e^{-rT} \frac{\partial}{\partial K} \left(- \int_K^\infty f_{S_T}(s) ds \right) = e^{-rT} f_{S_T}(K).$$

3.8 Explain the term of BSM

Provide a very detailed explanation of the meanings of $N(d_1)$ and $N(d_2)$ in the Black-Scholes European option pricing formula. Use these to derive the standard Black-Scholes European option pricing formula and ensure that the components of the formula match the standard interpretation.

Solution:

The Black-Scholes formula for the price of a European call option is given by:

$$c(t) = S(t)N(d_1) - e^{-r(T-t)}XN(d_2),$$

where:

- $S(t)$ is the current price of the underlying asset at time t ,
- X is the strike price of the option,
- r is the risk-free interest rate,
- T is the time to maturity of the option,
- $N(d_1)$ and $N(d_2)$ are the cumulative distribution functions (CDF) of the standard normal distribution evaluated at d_1 and d_2 , respectively.

The term $N(d_1)$ represents the "Delta" of the option, which is the sensitivity of the option price to changes in the underlying asset's price. This tells us the probability (in a risk-neutral world) that the option will finish in the money at expiration. $N(d_2)$ represents the risk-adjusted probability that the option will be exercised, adjusted by the factor $e^{-r(T-t)}$, which accounts for the present value of the strike price payment.

Derivation of the Formula:

The formula assumes that the underlying asset price follows a geometric Brownian motion, which leads to the Black-Scholes PDE. By solving this PDE with the appropriate boundary conditions, the solution for a European call option is given as above.

Another way to interpret $N(d_1)$ is as the "equivalent" number of shares the option holder should hold to replicate the option's payoff (Delta hedging). The term $N(d_2)$ gives the probability that the option will expire in the money, discounted to present value.

In particular, d_1 and d_2 are defined as follows:

$$d_1 = \frac{\ln\left(\frac{S(t)}{X}\right) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}},$$

$$d_2 = d_1 - \sigma\sqrt{T-t},$$

where σ is the volatility of the underlying asset.

Interpretation of $N(d_1)$ and $N(d_2)$

$N(d_1)$ represents the probability, adjusted for the option's risk, that the option will finish in the money. It is often referred to as the Delta of the option, which gives the sensitivity of the option price to small changes in the underlying asset price.

$N(d_2)$ represents the risk-neutral probability that the option will be exercised, or the likelihood that the asset price will be above the strike price at maturity. This term is discounted by $e^{-r(T-t)}$ to reflect the present value of receiving the strike price.

Put-Call Parity:

Call options and put options are related through the put-call parity relationship:

$$S(t) + p(t) = c(t) + Xe^{-r(T-t)} + D,$$

where D represents any dividends paid by the underlying asset. If dividends are zero, the equation simplifies to:

$$S(t) + p(t) = c(t) + Xe^{-r(T-t)}.$$

By substituting the formula for $c(t)$ and using the fact that $D = 0$, we derive the put option price as:

$$p(t) = e^{-r(T-t)}XN(-d_2) - S(t)N(-d_1),$$

which matches the above results.

3.9 Implied Volatility and Volatility Smile

Explain implied volatility and volatility smile. What is the implication of volatility smile for the Black-Scholes pricing model?

Solution:

Implied volatility is the volatility that makes the model option price equal to the market option price.

Volatility smile describes the relationship between the implied volatility of the options and the strike prices for a given asset.

For equity, volatility often decreases as the strike price increases (also called volatility skew).

The Black-Scholes model assumes that the asset price follows a lognormal distribution with constant volatility. In reality, volatilities are neither constant nor deterministic. In fact, the volatility is a stochastic process itself. Furthermore, there may be jumps in asset prices.

3.10 Alternative BSM

In the Black-Scholes model, calculate the value of a European option at time T with the following payoff:

$$\max(S_T^2 - K, 0),$$

where S_T is the price of the stock at time T and K is the strike price.

Solution:

In the Black-Scholes model, the stock price S_t follows a geometric Brownian motion described by the stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ is the drift rate, σ is the volatility, and W_t is a Wiener process (standard Brownian motion). Our task is to compute the expected value in the risk-neutral measure of the following:

$$\mathbb{E} [\max(S_T^2 - K, 0)].$$

Assuming constant interest rates r , the Black-Scholes formula uses the risk-neutral measure, where the drift rate μ is replaced by the risk-free rate r . Hence, the SDE becomes:

$$dS_t = r S_t dt + \sigma S_t dW_t.$$

To solve this, we need to express the stock price S_T^2 in terms of its distribution at time T . Applying Ito's Lemma to the geometric Brownian motion, we get the following expression for S_t :

$$S_t = S_0 \exp \left(\left(r - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right),$$

where S_0 is the stock price at $t = 0$. Now, we need to compute the expected value of S_T^2 . Using Ito's formula, we can compute the square of S_T as:

$$S_T^2 = S_0^2 \exp \left(2 \left(r - \sigma^2 \right) T + 2\sigma W_T \right).$$

By comparing this to the original geometric Brownian motion, we can conclude that under the risk-neutral valuation measure, S_t^2 follows a geometric Brownian motion with volatility 2σ and drift rate $2r - \sigma^2$.

Therefore, we can use the Black formula. We only need to write S_T^2 as:

$$F_T(0) e^{-\nu^2 T / 2 + \nu W_T}$$

Calculate the values of $F_T(0)$ and ν , then substitute into the formula for calculation.

4 Greeks

4.1 Delta

What is the delta of a European call and put option on a non-dividend paying stock? How do you derive the delta?

solution:

The delta of a European call on a non-dividend paying stock has a clean expression:

$$\Delta = N(d_1)$$

For a put option, we can use the put-call parity relationship, which gives:

$$P + S = C + e^{-r(T-t)}K$$

Therefore,

$$\frac{\partial P}{\partial S} = \frac{\partial C}{\partial S} - 1 = N(d_1) - 1.$$

For the derivation, though, many make the mistake by treating $N(d_1)$ and $N(d_2)$ as constants in the call pricing formula $c = SN(d_1) - Ke^{-rT}N(d_2)$ and simply taking the partial derivative on S to yield $N(d_1)$. The derivation step is actually more complex than that since both $N(d_1)$ and $N(d_2)$ are functions of S through d_1 and d_2 . So the correct partial derivative is

$$\frac{\partial c}{\partial S} = N(d_1) + S \times \frac{\partial}{\partial S} N(d_1) - Ke^{-rT} \frac{\partial}{\partial S} N(d_2)$$

Take the partial derivative with respect to S for $N(d_1)$ and $N(d_2)$:

$$\begin{aligned} \frac{\partial}{\partial S} N(d_1) &= N'(d_1) \frac{\partial}{\partial S} d_1 = \frac{1}{S\sigma\sqrt{\tau}} e^{-d_1^2/2} \times \frac{1}{\sqrt{2\pi}} = \frac{1}{S\sigma\sqrt{\tau}} e^{-d_1^2/2} \frac{1}{\sqrt{2\pi}} \\ \frac{\partial}{\partial S} N(d_2) &= N'(d_2) \frac{\partial}{\partial S} d_2 = \frac{1}{S\sigma\sqrt{\tau}} e^{-d_2^2/2} \times \frac{1}{\sqrt{2\pi}} = \frac{1}{S\sigma\sqrt{\tau}} e^{-(d_1 - \sigma\sqrt{\tau})^2/2} \end{aligned}$$

So we have

$$\frac{\partial}{\partial S} N(d_2) = \frac{S}{K} e^{rT} N(d_1) \Rightarrow S \times \frac{\partial}{\partial S} N(d_1) - Ke^{-rT} \frac{\partial}{\partial S} N(d_2) = 0$$

Hence, the last two components of $\frac{\partial c}{\partial S}$ cancel out and

$$\frac{\partial c}{\partial S} = N(d_1)$$

4.2 Delta Range

Delta's minimum and maximum values?

solution:

To simplify, assuming the underlying asset is a non-dividend-paying stock:

For a long call position in a European option, the Delta typically ranges between 0 and 1. Therefore, the Delta of a short call position generally falls between -1 and 0. The Delta of a long call position increases with the underlying asset's spot price. When the asset is worthless (i.e., the option is deep out-of-the-money), the Delta is 0. When the asset's spot price is extremely high (i.e., the option is deep in-the-money), the Delta is 1.

For a long put position in a European option, the Delta ranges between -1 and 0. Hence, the Delta of a short put position ranges between 0 and 1. The Delta of a long put position decreases with the underlying asset's spot price. When the asset is worthless (i.e., the option is deep in-the-money), the Delta is -1. When the asset's spot price is extremely high (i.e., the option is deep out-of-the-money), the Delta is 0.

Please note, based on the parity relationship between call and put options, $C - P = S - Ke^{-rT}$, we can derive that $\Delta(P) = \Delta(C) - 1$, which is consistent with the limits mentioned above.

4.3 Delta Value

What is the Delta of an at-the-money European call option? What is the Delta of an at-the-money European put option?

solution:

The Delta of an at-the-money European call option is approximately 0.5. The Delta of an at-the-money European put option is approximately -0.5.

Under the assumption of a zero risk-free interest rate and the underlying asset not paying dividends, in the Black-Scholes framework, the Delta of a European call option $\Delta(C_{BS}) = N(d_1)$, where $N(x)$ represents the cumulative distribution function of a standard normal variable, and:

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = \frac{\sigma\sqrt{T}}{2}$$

Given $K = S$ and $r = q = 0$.

For the linear Taylor approximation near 0 for $N(x)$:

$$N(x) \approx 0.5 + \frac{x}{\sqrt{2\pi}} \approx 0.5 + 0.4x,$$

since $\frac{1}{\sqrt{2\pi}} \approx 0.3989 \approx 0.4$. Thus,

$$\Delta(C_{BS}) = N(d_1) = N\left(\frac{\sigma\sqrt{T}}{2}\right) \approx 0.5 + 0.2\sigma\sqrt{T}.$$

This is often estimated as $\Delta(C_{BS}) \approx 0.5$ because for most options, $0.2\sigma\sqrt{T}$ is small (e.g., for a volatility less than 50

For European put options, $\Delta(P_{BS}) = -N(-d_1)$, where $d_1 = \frac{\sigma\sqrt{T}}{2}$:

$$\Delta(P_{BS}) = -N\left(\frac{\sigma\sqrt{T}}{2}\right) \approx -0.5 + 0.2\sigma\sqrt{T}.$$

Using the same reasoning, this is usually approximated as $\Delta(P_{BS}) \approx -0.5$.

4.4 Delta curve

What is your estimate of the delta of an at-the-money call on a stock without dividend? What will happen to delta as the at-the-money option approaches maturity date?

solution:

For an at-the-money European call, the stock price equals the strike price.

$$S = K \Rightarrow d_1 = \frac{(r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}} = \left(\frac{r}{\sigma} + \frac{\sigma}{2}\right)\sqrt{\tau} > 0$$

and

$$\Delta = N(d_1) > 0.5.$$

As shown in Figure, all at-the-money call options indeed have $\Delta > 0.5$,

and the longer the maturity, the higher the Δ . As $T - t \rightarrow 0$,

$$\left(\frac{r}{\sigma} + \frac{\sigma}{2}\right)\sqrt{\tau} \rightarrow 0 \Rightarrow N(d_1) = N(0) = 0.5,$$

which is also shown in Figure ($T = 10$ days). The same argument is true for calls on stock with continuous dividend rate y if $r > y$.

Figure also shows that when S is large ($S \gg K$), Δ approaches 1. Furthermore, the shorter the maturity, the faster the delta approaches 1. On the other hand, if S is small ($S \ll K$), Δ approaches 0, and the shorter the maturity, the faster the delta approaches 0.

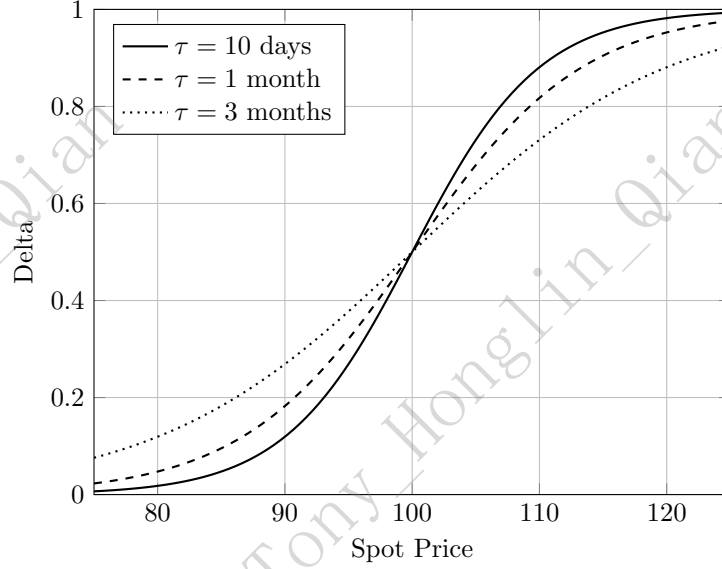


Figure 2: Delta of Call Options

4.5 Gamma

The Gamma of an option or option portfolio is the second derivative of the option value with respect to the underlying asset price:

$$\Gamma = \frac{\partial^2 V}{\partial S^2}.$$

According to the definition of Delta, Gamma is the first derivative of Delta, which is the sensitivity of Delta to the underlying asset price. It can be interpreted as the frequency and amount of adjustments needed to maintain a neutral Delta position. For a European call option, Gamma can be calculated as:

$$\frac{\partial^2 C}{\partial S^2} = \frac{\partial N(d_1)}{\partial S} = \frac{1}{\sqrt{2\pi}S\sigma\sqrt{T-t}}e^{-d_1^2/2}.$$

Therefore, the Gamma of a European call option is always positive. Thus, the price of a European call option is a convex function of the underlying asset price S . According to the principle of put call parity, **the Gamma of a call option is equal to the Gamma of a put option.**

4.6 Gamma curve

What happens to the gamma of an at-the-money European option when it approaches its maturity?

solution:

The gamma of options with shorter maturities approaches 0 much faster than options with longer maturities as S moves away from K . So for deep in-the-money or deep out-of-the-money options, longer maturity means higher gamma.

In contrast, if the stock prices are close to the strike price (at the money) as the maturity nears, the slope of delta for an at-the-money call becomes steeper and steeper. So for options close to the strike price, shorter-term options have higher gammas.

As $\tau \rightarrow 0$, an at-the-money call/put has $\Gamma \rightarrow \infty$ (Δ becomes a step function). This can be shown from the formula of gamma for a European call/put with no dividend:

$$\Gamma = \frac{N'(d_1)}{S\sigma\sqrt{\tau}}.$$

When $S = K$,

$$d_1 = \lim_{\tau \rightarrow 0} \left(\frac{r}{\sigma} + \frac{\sigma}{2} \right) \sqrt{\tau} \rightarrow 0 \Rightarrow \lim_{\tau \rightarrow 0} N'(d_1) = \frac{1}{\sqrt{2\pi}}.$$

The numerator is $\frac{1}{\sqrt{2\pi}}$, yet the denominator has a limit $\lim_{\tau \rightarrow 0} S\sigma\sqrt{\tau} \rightarrow 0$, so $\Gamma \rightarrow \infty$.

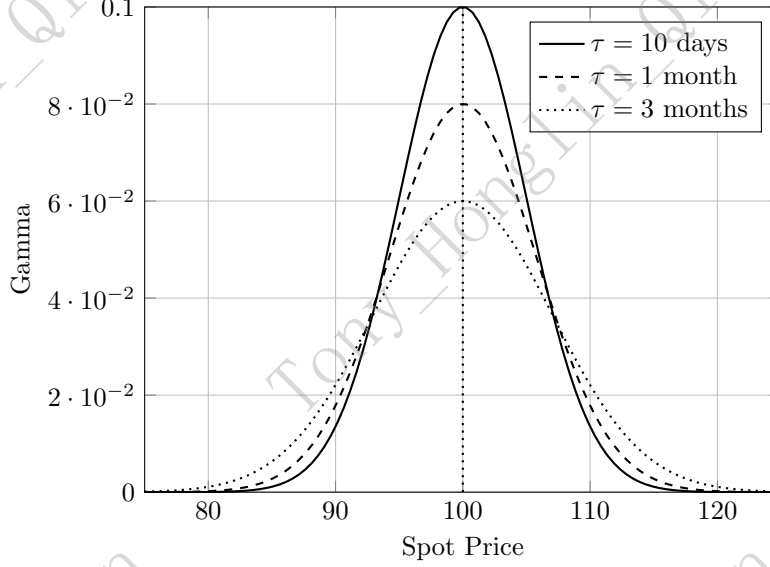


Figure 3: Variation of gamma of a European call option with respect to S and T . $K = 100$, $r = 0.05$, $\sigma = 0.25$.

4.7 Theta

The Theta of an option or option portfolio is the rate of change of the option price with respect to time:

$$\Theta = \frac{\partial V}{\partial t}$$

For a call option, we can use the following derivative to calculate its Theta:

$$\frac{\partial C}{\partial t} = -\frac{\sigma S}{2\sqrt{T-t}} \cdot \frac{1}{\sqrt{2\pi}} e^{-d_1^2/2} - rKe^{-r(T-t)}N(d_2)$$

Obviously, the Theta of a call option is always negative, so the option price decreases gradually over time. We have already learned that the intrinsic value of a call option diminishes as expiration approaches. Essentially, this is because the time value of a call option, similar to a put option, decreases as expiration approaches, leaving the option holder with less opportunity to exercise it. In trading, this time value decay is commonly referred to as the loss of time value.

For a put option, the situation is different. We can study its Theta using the put price relationship:

$$\frac{\partial P}{\partial t} = -\frac{\sigma S}{2\sqrt{T-t}} \cdot \frac{1}{\sqrt{2\pi}} e^{-d_1^2/2} - rKe^{-r(T-t)}N(-d_2) + rKe^{-r(T-t)}$$

So, the Theta of a put option is not always negative. When the underlying asset price is far below the strike price, Theta can be positive, meaning the price of the put option increases over time.

4.8 Theta Curve

When there is no dividend, the theta for a European call option is simplified to:

$$\Theta = -\frac{SN'(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}N(d_2)$$

which is always negative. As shown in Figure,

when $S \ll K$, $N(d_2) \approx 0$ and $N'(d_1) \approx 0$. Hence, $\Theta \rightarrow 0$.

When $S \gg K$, $N(d_2) \approx 1$ and $N'(d_1) \approx 0$. Hence, $\Theta \rightarrow -rKe^{-r\tau}$.

When $S \approx K$, Θ has a large negative value, and the smaller the τ , the more negative the Θ .

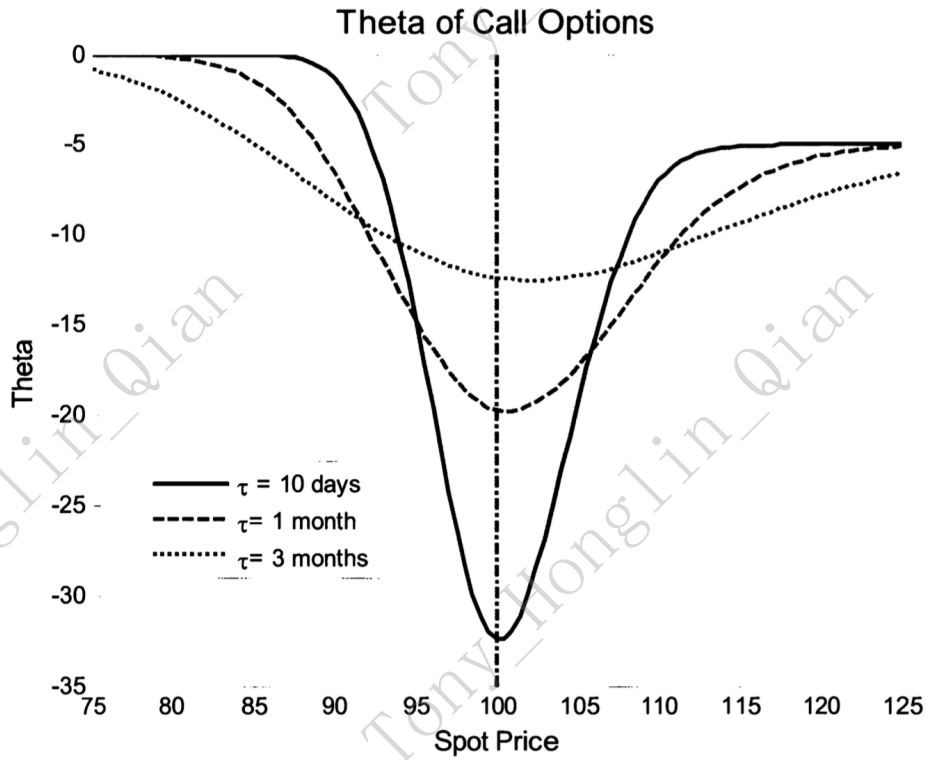


Figure 4: Theta Curve

4.9 Vega

The Vega of an option or an option portfolio is the sensitivity of the option price to changes in volatility:

$$\text{Vega} = \frac{\partial V}{\partial \sigma}$$

For a call option, the Vega is:

$$\frac{\partial C}{\partial \sigma} = \frac{S\sqrt{T-t}}{\sqrt{2\pi}} e^{-d_1^2/2}$$

Clearly, Vega is always positive. When volatility increases, the option price increases because future uncertainty increases. Options act like an insurance contract, where higher uncertainty results in higher

prices. The Vega of a put option is equal to that of a call option, so as volatility increases, the price of a put option also increases.

4.10 Rho

The Rho of an option or an option portfolio is the sensitivity of the option price to the risk-free interest rate in the Black-Scholes formula:

$$\rho = \frac{\partial V}{\partial r}$$

For call and put options, we have the following:

$$\begin{aligned}\frac{\partial C}{\partial r} &= K(T-t)e^{-r(T-t)}N(d_2), \\ \frac{\partial P}{\partial r} &= -K(T-t)e^{-r(T-t)}N(-d_2)\end{aligned}$$

Thus, when the interest rate increases, the price of a call option rises, while the price of a put option falls.

4.11 Call Option Greeks Summary

Question: Calculate the greeks $\Delta, \Gamma, \nu, \rho, \Theta$ for a call option.

Solution:

We denote C the price of the call option at $t = 0$, ϕ the standard normal cumulative distribution and $f = \phi'$ the standard normal density function.

$$C = S_0\phi(d_1) - Ke^{-rT}\phi(d_2)$$

Where,

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + rT + \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}}, \quad d_2 = \frac{\ln\left(\frac{S_0}{K}\right) + rT - \frac{\sigma^2 T}{2}}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

We start with an identity that will help us for all the greeks

$$\begin{aligned}f(d_2) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{d_2^2}{2}\right) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{d_1^2}{2}\right) \exp(d_1\sigma\sqrt{T}) \exp\left(-\frac{\sigma^2 T}{2}\right) \\ f(d_2) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{d_1^2}{2}\right) \frac{S_0}{K} \exp(rT) = f(d_1) \frac{S_0}{K} e^{rT} \quad (1)\end{aligned}$$

Δ :

$$\begin{aligned}\Delta &= \frac{\partial C}{\partial S_0} = \phi(d_1) + S_0 f(d_1) \frac{\partial d_1}{\partial S_0} - Ke^{-rT} f(d_2) \frac{\partial d_2}{\partial S_0} \\ \Delta &= \phi(d_1) + f(d_1) \frac{1}{\sigma\sqrt{T}} - Ke^{-rT} f(d_2) \frac{S_0}{\sigma\sqrt{T}}\end{aligned}$$

And using equation 1

$$\Delta = \phi(d_1)$$

Γ :

$$\Gamma = f(d_1) \frac{1}{S_0 \sigma \sqrt{T}}$$

ν :

$$\nu = \frac{\partial C}{\partial \sigma} = S_0 f(d_1) \frac{\partial d_1}{\partial \sigma} - K e^{-rT} f(d_2) \frac{\partial d_2}{\partial \sigma}$$

$$\nu = \frac{\partial C}{\partial \sigma} = S_0 f(d_1) \frac{\partial d_1}{\partial \sigma} - K e^{-rT} f(d_2) \left(\frac{\partial d_1}{\partial \sigma} - \sqrt{T} \right)$$

And using equation 1

$$\nu = K e^{-rT} f(d_2) \sqrt{T} = S_0 f(d_1) \sqrt{T}$$

ρ :

$$\rho = \frac{\partial C}{\partial r} = S_0 f(d_1) \frac{\partial d_1}{\partial r} + T K e^{-rT} \phi(d_2) - K e^{-rT} f(d_2) \frac{\partial d_2}{\partial r}$$

$$\rho = S_0 f(d_1) \frac{\sqrt{T}}{\sigma} + T K e^{-rT} \phi(d_2)$$

And using equation 1

$$\rho = T K e^{-rT} \phi(d_2)$$

Θ :

$$\Theta = \frac{\partial C}{\partial T} = S_0 f(d_1) \frac{\partial d_1}{\partial T} + r K e^{-rT} \phi(d_2) - K e^{-rT} f(d_2) \frac{\partial d_2}{\partial T}$$

$$\Theta = S_0 f(d_1) \frac{\partial d_1}{\partial T} + r K e^{-rT} \phi(d_2) - K e^{-rT} f(d_2) \left(\frac{\partial d_1}{\partial T} - \frac{\sigma}{2\sqrt{T}} \right)$$

And using equation 1

$$\Theta = r K e^{-rT} \phi(d_2) + K e^{-rT} f(d_2) \frac{\sigma}{2\sqrt{T}}$$

5 Option Trading

5.1 Spread

Spread Type	Market Outlook	Options Involved	Profit/Loss Potential
Bull Call Spread	Bullish	Buy call (lower strike), sell call (higher strike)	Limited profit, limited loss
Bull Put Spread	Bullish	Sell put (higher strike), buy put (lower strike)	Limited profit, limited loss
Bear Call Spread	Bearish	Sell call (lower strike), buy call (higher strike)	Limited profit, limited loss
Bear Put Spread	Bearish	Buy put (higher strike), sell put (lower strike)	Limited profit, limited loss
Butterfly Call Spread	Neutral	Buy 1 call (lower strike), sell 2 calls (middle strike), buy 1 call (higher strike)	Limited profit, limited loss
Butterfly Put Spread	Neutral	Buy 1 put (lower strike), sell 2 puts (middle strike), buy 1 put (higher strike)	Limited profit, limited loss
Calendar Call Spread	Neutral / Volatility	Buy call (longer expiry), sell call (shorter expiry)	Profit from time decay, limited risk
Calendar Put Spread	Neutral / Volatility	Buy put (longer expiry), sell put (shorter expiry)	Profit from time decay, limited risk

Table 1: Options Trading Spread Strategies

5.2 Combination

Strategy	Market Outlook	Options Involved	Profit/Loss Potential
Long Straddle	Volatile (expecting large move in either direction)	Buy 1 call (ATM), Buy 1 put (ATM)	Unlimited profit, limited loss (total premium paid)
Short Straddle	Stable (expecting little movement)	Sell 1 call (ATM), Sell 1 put (ATM)	Limited profit (premium received), unlimited risk
Long Strangle	Volatile (expecting large move in either direction)	Buy 1 OTM call, Buy 1 OTM put	Unlimited profit, limited loss (total premium paid)
Short Strangle	Stable (expecting little movement)	Sell 1 OTM call, Sell 1 OTM put	Limited profit (premium received), unlimited risk

Table 2: Combination Strategies in Options Trading

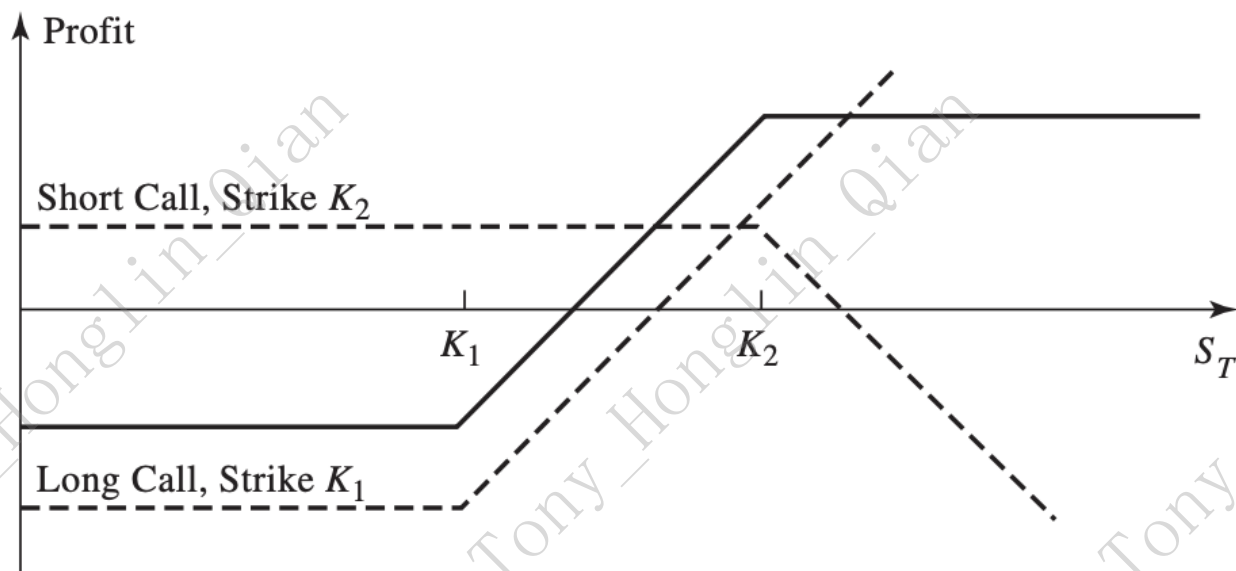


Figure 5: Bull Spread Call

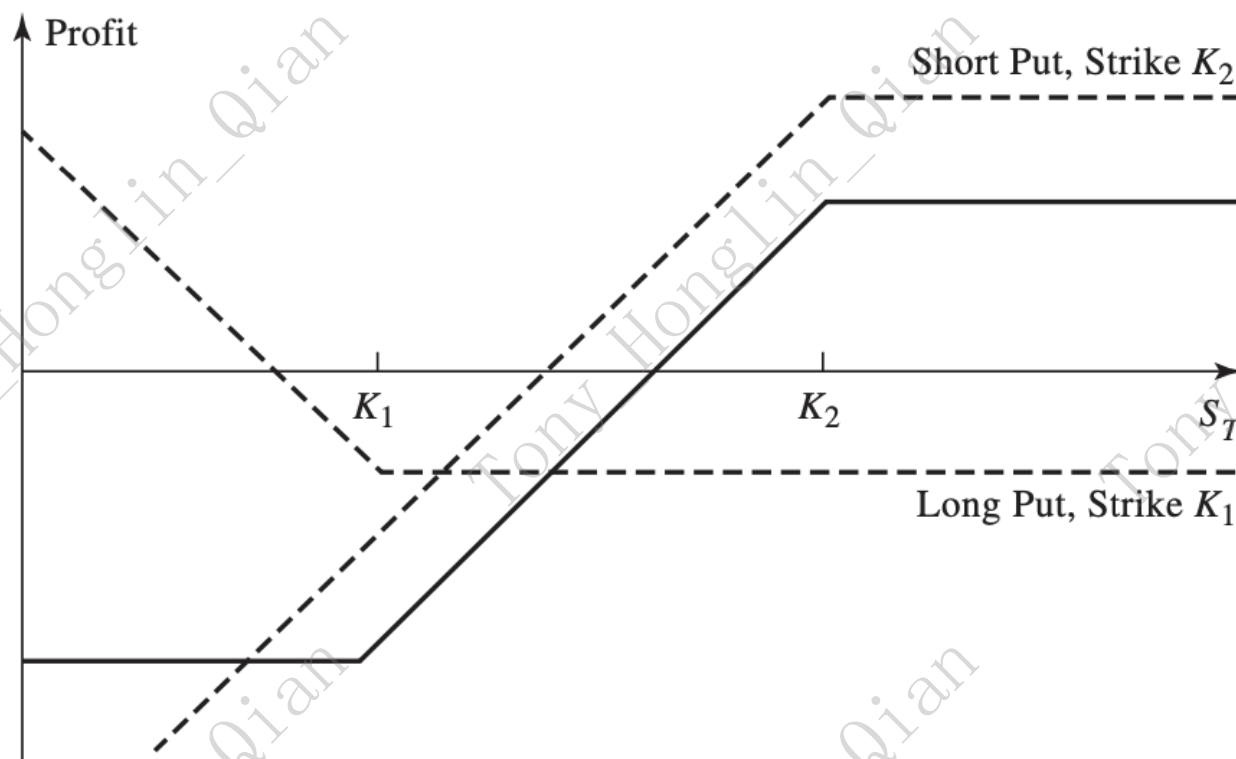


Figure 6: Bull Spread put

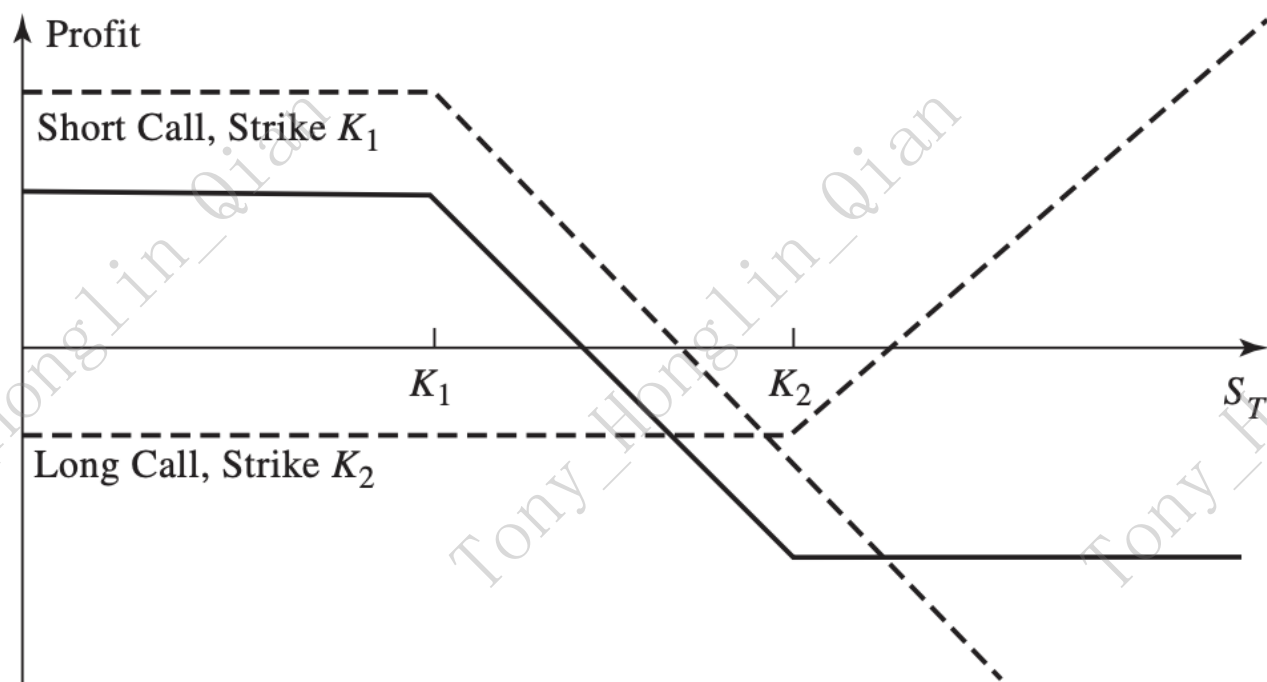


Figure 7: Bear Spread Call

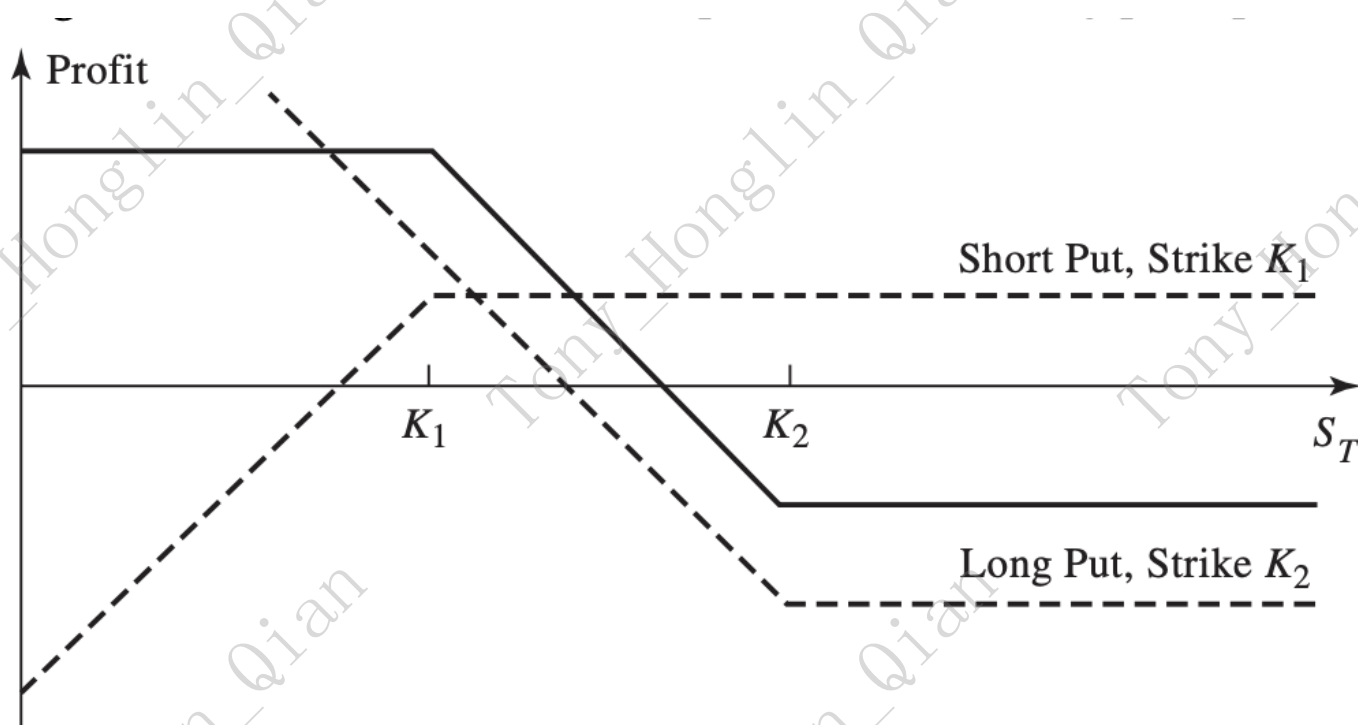


Figure 8: Bear Spread Put

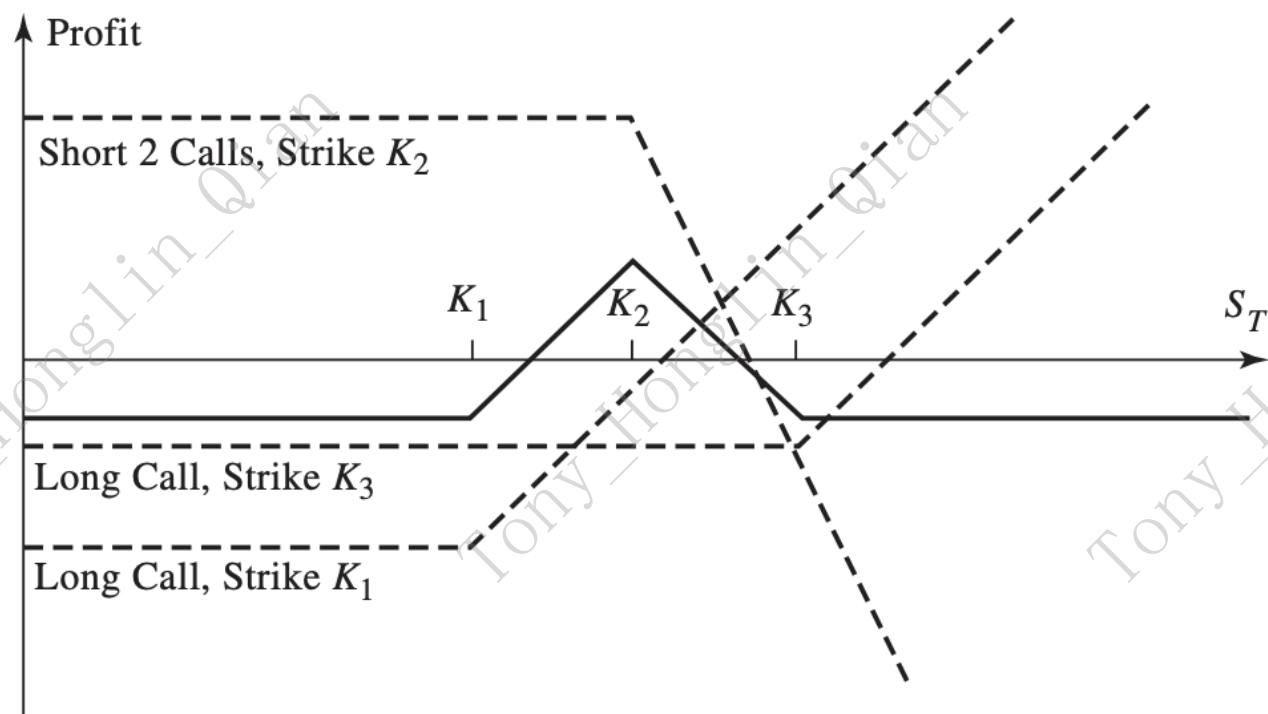


Figure 9: Butterfly Call

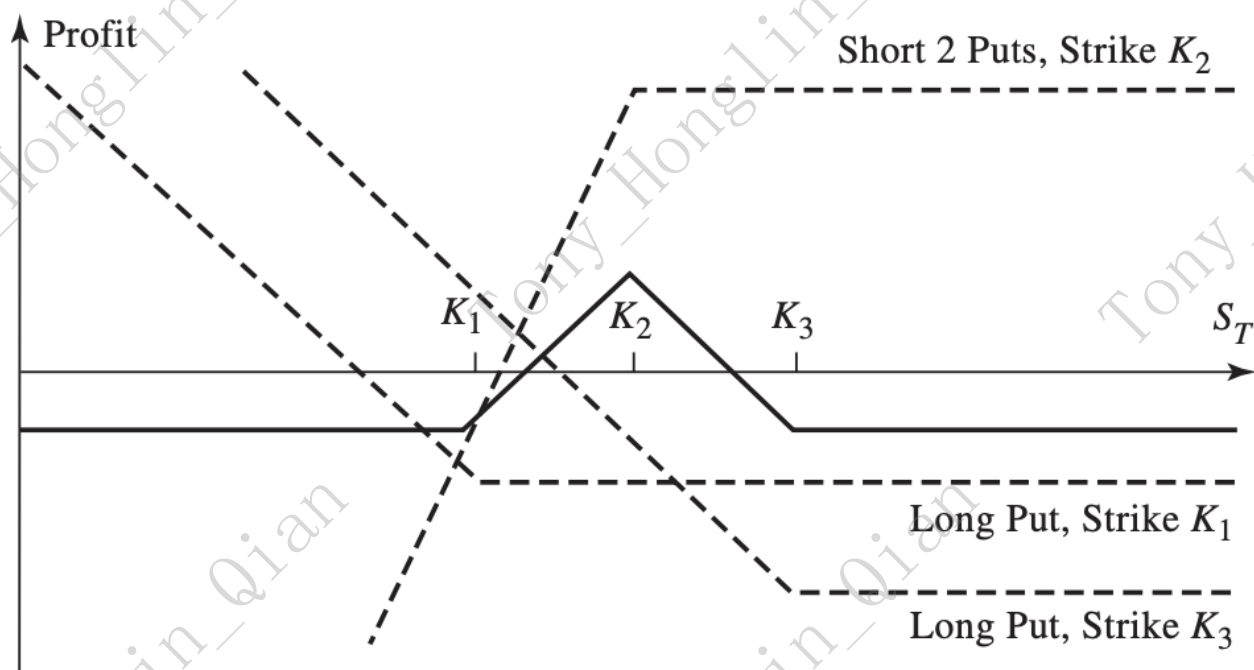


Figure 10: Butterfly Put

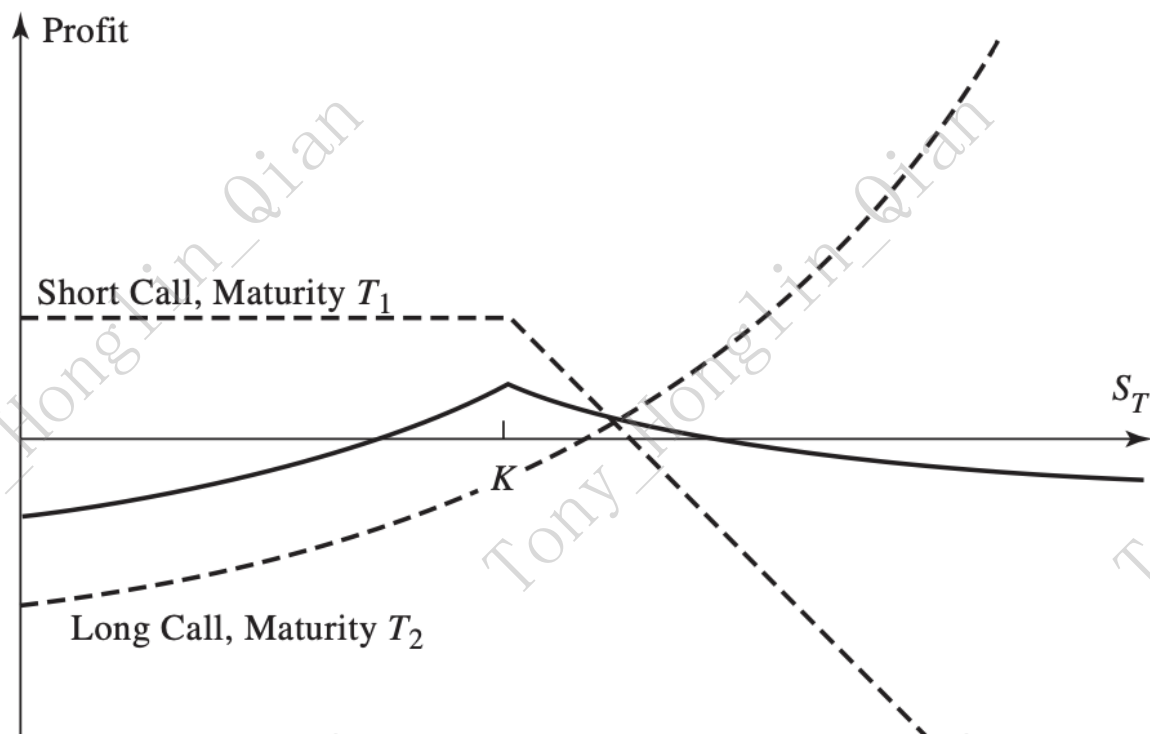


Figure 11: Calendar Call

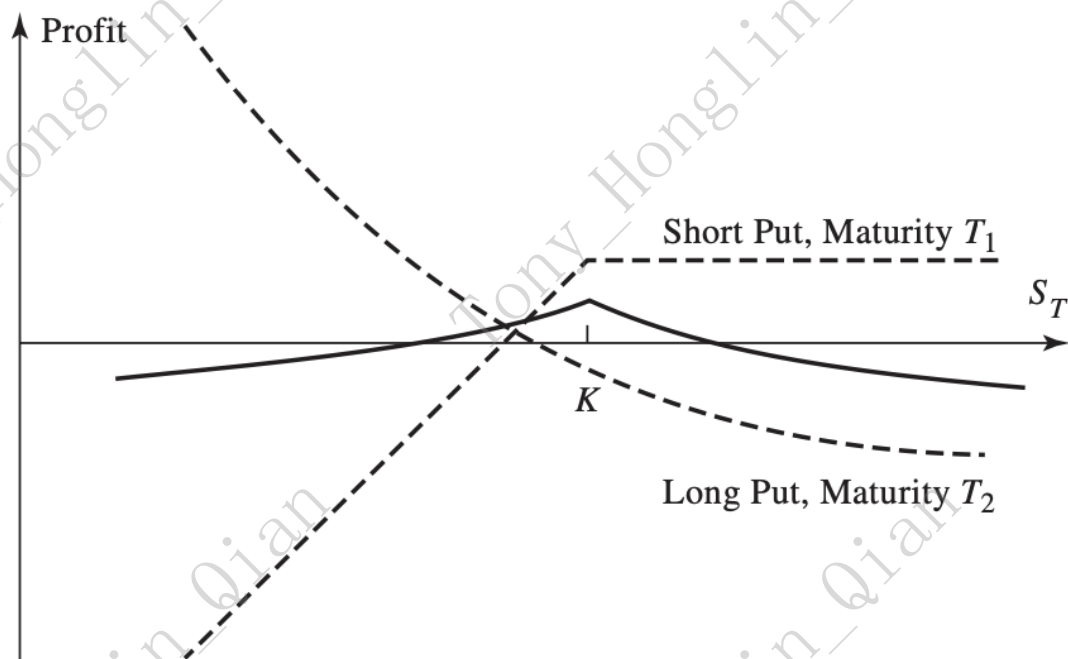


Figure 12: Calendar Put

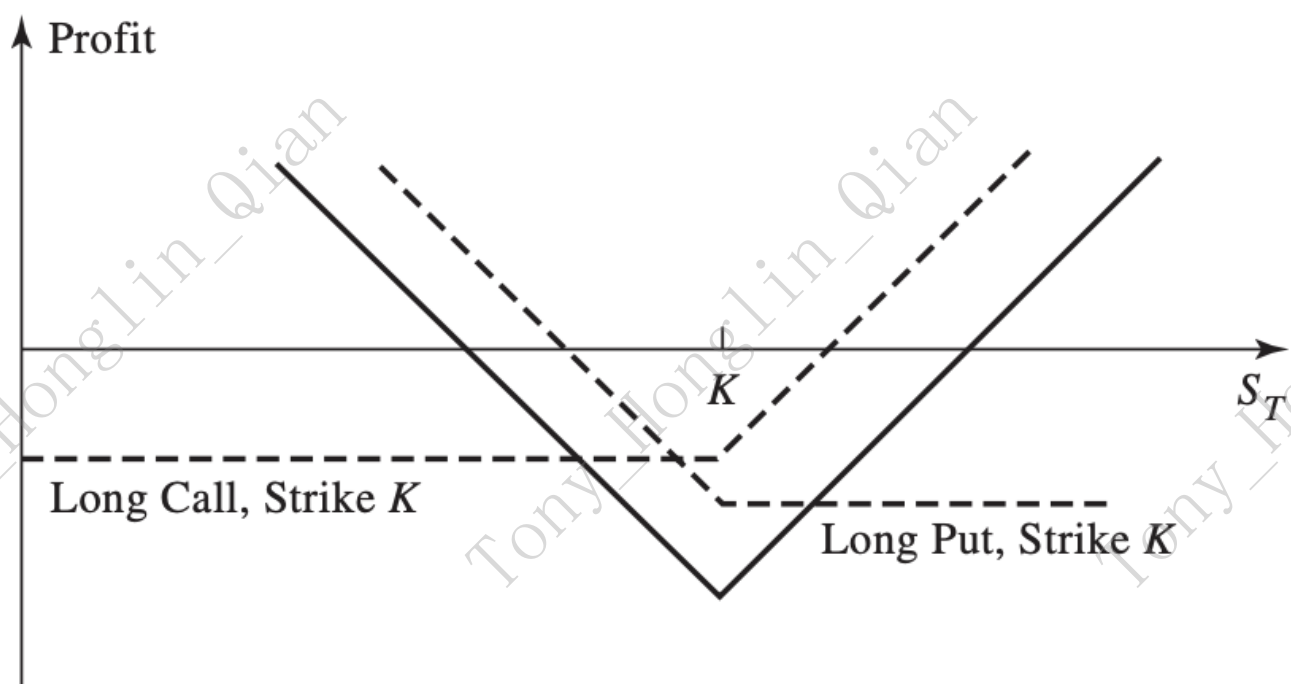


Figure 13: Long Straddle

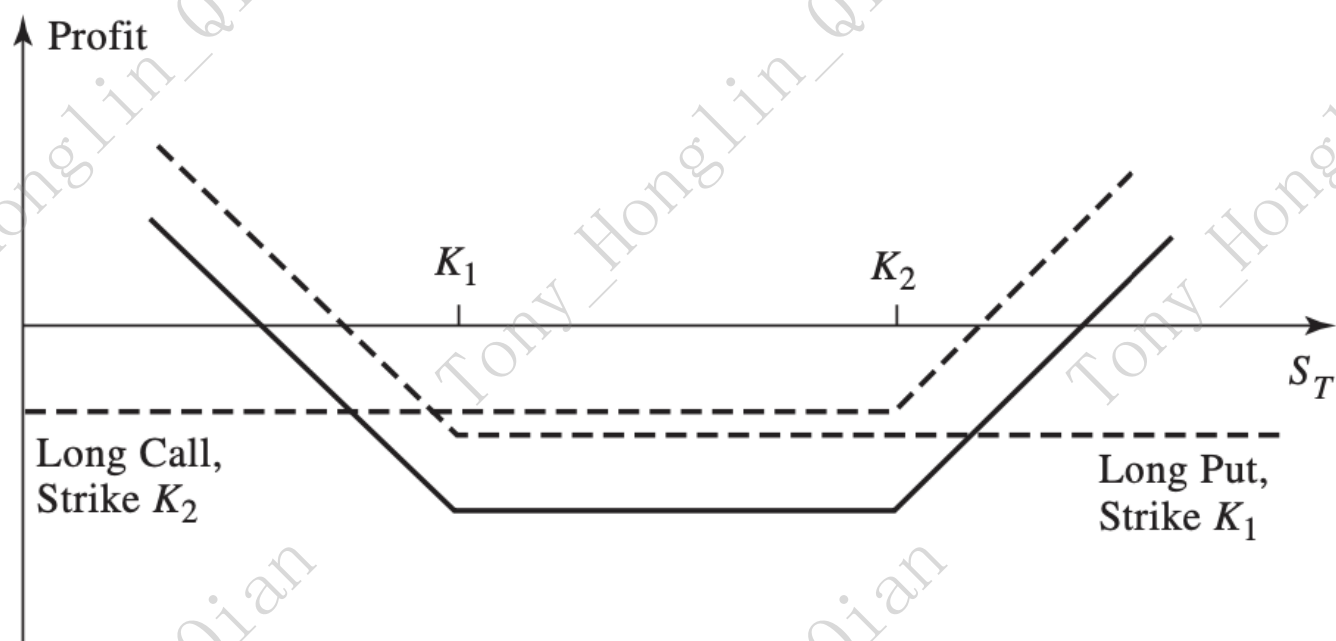


Figure 14: Long Strangle

6 Basic Numerical Method

6.1 Binomial Tree

1. Asset Price Evolution

The up and down factors are given by:

$$u = e^{\sigma\sqrt{\Delta t}} \quad (1)$$

$$d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}} \quad (2)$$

where σ is the volatility of the asset, and $\Delta t = \frac{T}{N}$ is the length of each time step, with T as the time to expiration and N as the number of time steps.

2. Risk-neutral Probability

The risk-neutral probability is:

$$p = \frac{e^{r\Delta t} - d}{u - d} \quad (3)$$

where r is the risk-free interest rate, and Δt is the time step.

The probability of a downward movement is $1 - p$.

3. Asset Price Formula

The price of the asset at time step i and node j is given by:

$$S_i = S_0 \cdot u^j \cdot d^{i-j} \quad (4)$$

where:

- S_0 is the initial price of the underlying asset.
- $u = e^{\sigma\sqrt{\Delta t}}$ is the up factor.
- $d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$ is the down factor.
- σ is the volatility of the underlying asset.
- $\Delta t = \frac{T}{N}$ is the time step size.
- i is the time step index.
- j is the number of up moves in the i -th time step.

4. American Call Option Pricing

At expiration $t = T$, the option value is:

$$C_N = \max(S_N - K, 0) \quad (5)$$

where S_N is the asset price at expiration and K is the strike price.

Working backward, the option value at each previous node i is:

$$C_i = \max(S_i - K, e^{-r\Delta t} (pC_{i+1}^{up} + (1-p)C_{i+1}^{down})) \quad (6)$$

where C_{i+1}^{up} and C_{i+1}^{down} are the option values after an upward and downward movement, respectively.

5. American Put Option Pricing

At expiration $t = T$, the option value is:

$$P_N = \max(K - S_N, 0) \quad (7)$$

Working backward, the option value at each previous node i is:

$$P_i = \max(K - S_i, e^{-r\Delta t} (pP_{i+1}^{up} + (1-p)P_{i+1}^{down})) \quad (8)$$

6. Parameters Explanation

- S : Current price of the underlying asset.
- K : Strike price of the option.
- T : Time to expiration.
- r : Risk-free interest rate.
- σ : Volatility of the underlying asset.
- N : Number of time steps in the binomial tree.
- Δt : Length of each time step.
- u : Upward movement factor for the asset price.
- d : Downward movement factor for the asset price.
- p : Risk-neutral probability of an upward price movement.
- C_i, P_i : Option price at node i for call and put options, respectively.

6.2 Monte Carlo Simulation for Option Pricing

1. Geometric Brownian Motion (GBM) Model

The asset price S_t follows a stochastic differential equation described by Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where:

- S_t is the asset price at time t ,
- μ is the drift rate (expected return),
- σ is the volatility of the asset,
- dW_t is a Wiener process (random shock).

For the discrete case, the asset price evolves according to:

$$S_{t+\Delta t} = S_t \cdot e^{(\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}Z}$$

where $Z \sim N(0,1)$ is a standard normal random variable.

2. Monte Carlo Simulation Algorithm

The Monte Carlo algorithm for option pricing involves the following steps:

1. Simulate N possible price paths for the underlying asset using the GBM model.
2. For each simulated path, calculate the payoff at the option's maturity T . For a European call option, the payoff is:

$$\text{Payoff}_i = \max(S_T - K, 0)$$

where S_T is the simulated asset price at maturity, and K is the strike price.

3. Average the payoffs over all simulated paths:

$$\text{Average Payoff} = \frac{1}{N} \sum_{i=1}^N \text{Payoff}_i$$

4. Discount the average payoff back to the present value using the risk-free rate r :

$$\text{Option Price} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^N \text{Payoff}_i$$

3. Parameters

- S_0 : Initial price of the underlying asset.
- μ : Expected drift rate (or return) of the asset.
- σ : Volatility of the asset.
- r : Risk-free interest rate.
- T : Time to expiration.
- K : Strike price of the option.
- N : Number of simulations.

4. Example: European Call Option Pricing

For a European call option, the payoff at maturity T is given by:

$$\text{Payoff}_i = \max(S_T - K, 0)$$

The Monte Carlo estimate of the option price is:

$$\text{Option Price} = e^{-rT} \cdot \frac{1}{N} \sum_{i=1}^N \text{Payoff}_i$$

6.3 Finite Difference

6.3.1 Implicit Finite Difference Method

For an interior point (i, j) on the grid, the partial derivative of f with respect to S can be approximated as:

$$\frac{\partial f}{\partial S} = \frac{f_{i,j+1} - f_{i,j}}{\Delta S}$$

or as

$$\frac{\partial f}{\partial S} = \frac{f_{i,j} - f_{i,j-1}}{\Delta S}$$

We use a symmetrical approximation by averaging the two:

$$\frac{\partial f}{\partial S} = \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta S}$$

For the time derivative, we use a forward difference approximation so that the value at time $i\Delta t$ is related to the value at time $(i+1)\Delta t$:

$$\frac{\partial f}{\partial t} = \frac{f_{i+1,j} - f_{i,j}}{\Delta t}$$

For the second derivative $\frac{\partial^2 f}{\partial S^2}$, the backward difference approximation for $\frac{\partial f}{\partial S}$ at the (i, j) point is:

$$\frac{f_{i,j+1} - f_{i,j}}{\Delta S}$$

The backward difference at the $(i, j+1)$ point is:

$$\frac{f_{i,j+1} - f_{i,j}}{\Delta S}$$

Hence a finite difference approximation for $\frac{\partial^2 f}{\partial S^2}$ at the (i, j) point is:

$$\frac{\partial^2 f}{\partial S^2} = \frac{(f_{i,j+1} - f_{i,j}) - (f_{i,j} - f_{i,j-1})}{\Delta S}$$

Simplified, we get:

$$\frac{\partial^2 f}{\partial S^2} = \frac{f_{i,j+1} + f_{i,j-1} - 2f_{i,j}}{\Delta S^2}$$

Substituting equations into the BSM differential equation:

$$\frac{f_{i+1,j} - f_{i,j}}{\Delta t} + (r - q)j\Delta S \frac{f_{i,j+1} - f_{i,j-1}}{2\Delta S} + \frac{1}{2}\sigma^2 j^2 \Delta S^2 \frac{f_{i,j+1} + f_{i,j-1} - 2f_{i,j}}{\Delta S^2} = rf_{i,j}$$

Rearranging terms, we obtain:

$$a_j f_{i,j-1} + b_j f_{i,j} + c_j f_{i,j+1} = f_{i+1,j}$$

Where:

$$a_j = \frac{1}{2}(r - q)j\Delta t - \frac{1}{2}\sigma^2 j^2 \Delta t$$

$$b_j = 1 + \sigma^2 j^2 \Delta t + r\Delta t$$

$$c_j = -\frac{1}{2}(r - q)j\Delta t + \frac{1}{2}\sigma^2 j^2 \Delta t$$

Boundary Condition

The value of the put at time T is:

$$f_{N,j} = \max(K - S_T, 0)$$

Hence, the final value at the terminal node is:

$$f_{N,j} = \max(K - j\Delta S, 0), \quad j = 0, 1, \dots, M$$

The value of the put option when the stock price is zero is:

$$f_{i,0} = K, \quad i = 0, 1, \dots, N$$

We assume that the put option is worth zero when $S = S_{\max}$:

$$f_{i,M} = 0, \quad i = 0, 1, \dots, N$$

Using equation with $i = N - 1$, we get:

$$a_j f_{N-1,j-1} + b_j f_{N-1,j} + c_j f_{N-1,j+1} = f_{N,j}$$

6.3.2 Explicit Finite Difference Method

The implicit finite difference method has the advantage of being robust and always converging to the solution as ΔS and Δt approach zero. However, the disadvantage is that $M - 1$ simultaneous equations must be solved. The explicit method can simplify this by assuming the values of $\frac{\partial f}{\partial S}$ and $\frac{\partial^2 f}{\partial S^2}$ at point (i, j) are the same as at point $(i + 1, j)$.

The equations become:

$$\begin{aligned} \frac{\partial f}{\partial S} &= \frac{f_{i+1,j+1} - f_{i+1,j-1}}{2\Delta S} \\ \frac{\partial^2 f}{\partial S^2} &= \frac{f_{i+1,j+1} + f_{i+1,j-1} - 2f_{i+1,j}}{\Delta S^2} \end{aligned}$$

The difference equation is:

$$\frac{f_{i+1,j} - f_{i,j}}{\Delta t} + (r - q)j\Delta S \frac{f_{i+1,j+1} - f_{i+1,j-1}}{2\Delta S} + \frac{1}{2}\sigma^2 j^2 \Delta S^2 \frac{f_{i+1,j+1} + f_{i+1,j-1} - 2f_{i+1,j}}{\Delta S^2} = r f_{i,j}$$

Or:

$$f_{i,j} = a_j^* f_{i+1,j-1} + b_j^* f_{i+1,j} + c_j^* f_{i+1,j+1}$$

Where:

$$\begin{aligned} a_j^* &= \frac{1}{1 + r\Delta t} \left(-\frac{1}{2}(r - q)j\Delta t + \frac{1}{2}\sigma^2 j^2 \Delta t \right) \\ b_j^* &= \frac{1}{1 + r\Delta t} (1 - \sigma^2 j^2 \Delta t) \\ c_j^* &= \frac{1}{1 + r\Delta t} \left(\frac{1}{2}(r - q)j\Delta t + \frac{1}{2}\sigma^2 j^2 \Delta t \right) \end{aligned}$$

This creates what is known as the explicit finite difference method.

Difference:

The difference between the implicit and explicit methods is as follows:

The implicit method leads to equation, which gives a relationship between three different values of the option at time $i\Delta t$ (i.e., $f_{i,j-1}$, $f_{i,j}$, and $f_{i,j+1}$) and one value of the option at time $(i + 1)\Delta t$ (i.e., $f_{i+1,j}$).

The explicit method leads to equation, which gives a relationship between one value of the option at time $i\Delta t$ (i.e., $f_{i,j}$) and three different values of the option at time $(i + 1)\Delta t$ (i.e., $f_{i+1,j-1}$, $f_{i+1,j}$, and $f_{i+1,j+1}$).

6.4 Local Volatility

Question: Derive the Dupire Formula or Local Volatility.

Solution:

It might be surprising but this question is sometimes asked in interview. Local volatility is such an industry standard that you absolutely need to know it after a few years as a quant. Remember that the local volatility $\sigma_{loc}(t, S_t)$ when injected in the process S_t

$$\frac{dS_t}{S_t} = (r(t) - q(t)) dt + \sigma_{loc}(t, S_t) dW_t$$

matches the vanilla prices, or in other terms, the implied volatility surface, because option prices are actually the implied volatility surface. Dupire proved the existence and uniqueness of the local volatility surface in his seminal 1994 paper. The idea is to connect the stock dynamics to the stock distribution using the Fokker Planck equation, and then to the call prices using the Breeden-Litzenberger formula.

So we start by assuming that the stock prices follow

$$\frac{dS_t}{S_t} = (r(t) - q(t)) dt + \sigma(t, S_t) dW_t$$

So the Breeden-Litzenberger formula between times t and T , for a stock of spot value x , at time t , and discounting factor $D(t, T)$ (typically $D(t, T) = e^{-r(T-t)}$) is

$$p(s, K, t, T) = \frac{1}{D(t, T)} \frac{\partial^2 C(s, K, t, T)}{\partial K^2}$$

We apply Fokker-Planck to the stock

$$\begin{aligned} & \frac{1}{2} \frac{\partial^2}{\partial x^2} [\sigma(t, x)^2 x^2 p(x_0, x, t_0, t)] - (r(t) - q(t)) \frac{\partial}{\partial x} [xp(x_0, x, t_0, t)] \\ & - \frac{\partial}{\partial t} p(x_0, x, t_0, t) = 0 \end{aligned}$$

We multiply this by $D(t_0, t)(x = K)^+$ and integrate from $x = K$ to $x = \infty$ to get

$$\begin{aligned} & \frac{1}{2} D(t_0, t) \left[\frac{\partial^2}{\partial x^2} [\sigma(t, x)^2 x^2 p(x_0, x, t_0, t)]_{x=K} (x - K) dx \right] \\ & - (r(t) - q(t)) D(t_0, t) \left[\frac{\partial}{\partial x} [xp(x_0, x, t_0, t)]_{x=K} (x - K) dx \right] \quad (7) \end{aligned}$$

The first term in equation (7) can be integrated by parts and using the Breeden-Litzenberger formula it can be rewritten as

$$\frac{1}{2} \sigma(t, K)^2 K^2 \frac{\partial^2}{\partial K^2} C_{t_0}(x_0, K, t) \quad (8)$$

The second term in equation (7) can be integrated by parts and using the integral version of the Breeden-Litzenberger formula it becomes

$$(r(t) - q(t)) \left[C_{t_0}(x_0, K, t) - K \frac{\partial}{\partial K} C_{t_0}(x_0, K, t) \right] \quad (9)$$

The last term in equation (7) can be integrated directly and using

$$D(t_0, t) = -r(t) D(t_0, t)$$

we obtain

$$- \left(r(t)C(x_t, K, t) + \frac{\partial}{\partial t} C_{t_0}(x_0, K, t) \right) \quad (10)$$

After substituting the three terms and rearranging equation (7) we obtain the Dupire Formula

$$\sigma(t, K)^2 = \frac{(r(t) - q(t)) K \frac{\partial}{\partial K} C_{t_0}(x_0, K, t) + \frac{\partial}{\partial t} C_{t_0}(x_0, K, t) + q(t)C(x_0, K, t)}{\frac{1}{2} K^2 \frac{\partial^2}{\partial K^2} C_{t_0}(x_0, K, t)}$$

Usually we find it written in this compact form

$$\sigma_{loc}(t, K)^2 = \frac{\frac{\partial C}{\partial t} + (r(t) - q(t)) K \frac{\partial C}{\partial K} + q(t)C}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}$$

$$\sigma_{loc}(t, K)^2 = \frac{C_t + (r - q)KC_K + qC}{\frac{1}{2}K^2 C_{KK}}$$

6.5 Heston model

What is the Heston model?

Answer:

In Heston's stochastic volatility model, it is assumed that the price of the underlying asset satisfies the same SDE as in the lognormal model, i.e.,

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t,$$

whereas the instantaneous variance v_t itself follows a mean reverting CIR (Cox-Ingersoll-Ross) process

$$dv_t = \lambda(v_t - m)dt + \eta\sqrt{v_t}dZ_t,$$

where $\lambda > 0$ and $m > 0$ are positive constants. The driving Wiener processes W_t and Z_t are correlated with constant correlation ρ , i.e.,

$$\text{corr}(dW_t, dZ_t) = \rho dt.$$

The Heston model is a benchmark, and is commonly used in derivative pricing because it has the following features:

- it takes into account the leverage effect, namely, the driving Wiener processes W_t and Z_t are correlated; empirically, the correlation ρ is negative and that is why the word "leverage";
- it has a quasi closed form (up to an inverse Fourier transform) solution for the prices of European options, which make the calibration more tractable;
- the variance process is mean reverting with rate of reversion λ and long term mean m .

Note that, if the parameters of the volatility process are in the regime $2\lambda m < \eta^2$, then zero is an attainable boundary for the volatility process. Practitioners usually assume that the boundary behavior at zero is either absorption, i.e., the process is stuck at zero once it hits zero, or reflection, i.e., the process bounces back right after it hits zero. On the other hand, the other boundary infinity is an unattainable boundary.

7 Value-at-risk

7.1 VaR

Value at Risk (VaR) Calculation Methods with advantages and disadvantages

solution:

Value at Risk (VaR) is a widely used risk management tool that quantifies the potential loss in value of a portfolio or investment over a defined period, under normal market conditions, at a given confidence level. There are several methods for calculating VaR, each with its own advantages and disadvantages. Below are three common approaches:

1. Historical Simulation VaR

Historical simulation involves using past market data to simulate portfolio returns and calculate VaR by determining the percentile of losses over a given period.

Advantages:

- Simple to implement as it uses actual historical data.
- Reflects real market conditions and extreme market movements.
- Suitable for portfolios with non-linear instruments like options.

Disadvantages:

- Highly dependent on historical data; may not capture future extreme events.
- Assumes future market behavior will be similar to the past.
- Limited by the length and quality of available data.

If X_i represents the ordered historical returns, and we have N historical observations, the VaR at confidence level α is calculated as:

$$\text{VaR}_\alpha = X_{[\alpha \cdot N]}$$

Where:

- α is the confidence level (e.g., 0.05 for 5% VaR),
- $[\cdot]$ denotes the floor function, which rounds down to the nearest integer,
- N is the total number of historical observations,
- X_i are the ordered historical losses.

2. Parametric (Variance-Covariance) VaR

This method assumes that asset returns follow a normal distribution and uses the portfolio's mean and standard deviation to calculate VaR.

Advantages:

- Simple and computationally efficient to calculate.
- Quick and suitable for routine risk assessments.
- Easy to implement for linear portfolios.

Disadvantages:

- Assumes returns follow a normal distribution, which may not capture fat tails or skewness.

- Not suitable for portfolios with non-linear instruments like options.
- Underestimates risk during periods of market turmoil or crashes.

The parametric VaR formula assumes that returns follow a normal distribution. For a portfolio with normally distributed returns, the VaR is given by:

$$\text{VaR}_\alpha = \mu + Z_\alpha \cdot \sigma$$

Where:

- μ is the expected return of the portfolio,
- σ is the standard deviation (volatility) of the portfolio returns,
- Z_α is the critical value from the standard normal distribution corresponding to the confidence level α (e.g., $Z_{0.05} = -1.645$ for a 95% confidence level, $Z_{0.01} = -2.33$ for a 99% confidence level).

For simplicity, if the expected return μ is assumed to be 0 (as is often the case for short-term risk measures), the formula simplifies to:

$$\text{VaR}_\alpha = Z_\alpha \cdot \sigma$$

3. Monte Carlo Simulation VaR

Monte Carlo simulation generates random price paths based on statistical models to estimate potential losses and calculate VaR.

Advantages:

- Flexible and does not assume normal distribution of returns.
- Can model complex portfolios and non-linear instruments.
- Capable of incorporating different market scenarios for stress testing.

Disadvantages:

- Computationally intensive, especially for large portfolios.
- Requires careful modeling assumptions to avoid inaccuracies.
- More complex to implement and maintain.

The Monte Carlo VaR method does not have a single formula but rather follows a stochastic simulation process:

1. Assume a distribution for the portfolio returns (e.g., normal distribution).
2. Simulate a large number M of possible future price paths based on the statistical properties (mean μ , volatility σ , and correlation) of the assets in the portfolio.
3. Calculate the portfolio returns for each simulated path.
4. Sort the simulated returns and determine the percentile corresponding to the chosen confidence level α .

Mathematically, if $\tilde{X}_i$ represents the simulated portfolio returns, and M is the number of simulations, the VaR at confidence level α is calculated as:

$$\text{VaR}_\alpha = \tilde{X}_{[\alpha \cdot M]}$$

Where:

- α is the confidence level (e.g., 0.05 for 5% VaR),
- $\tilde{X}_i$ are the ordered simulated portfolio returns,
- $[\cdot]$ denotes the floor function, which rounds down to the nearest integer.

Method	Advantages	Disadvantages
Historical	Simple, real data, no distribution assumptions	Limited by past data, assumes future following past
Parametric	Fast, easy, good for linear portfolios	Assumes normal distribution, not for non-linear risk
Monte Carlo	Flexible, non-linear, stress-test capable	Computationally intensive, complex modeling

7.2 Portfolio VaR

Formula for Portfolio VaR

The formula for calculating the portfolio VaR for a portfolio consisting of two assets is given by:

$$\text{VaR}_p^2 = z^2 (w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho)$$

Where:

- z is the critical value corresponding to the desired confidence level (e.g., 1.645 for 95%, 2.33 for 99%),
- w_1 and w_2 are the portfolio weights of asset 1 and asset 2,
- σ_1^2 and σ_2^2 are the variances (or volatilities) of asset 1 and asset 2,
- ρ is the correlation between the returns of the two assets.

This formula takes into account both the individual variances of the assets and their correlation to calculate the overall risk of the portfolio.

Example Calculation

Suppose we have the following parameters:

- $\text{VaR}_1 = 10$,
- $\text{VaR}_2 = 10$,
- $w_1 = \frac{1}{2}$,
- $w_2 = \frac{1}{2}$,
- ρ

The portfolio VaR can be calculated using the following formula:

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 \cdot w_1^2 + \text{VaR}_2^2 \cdot w_2^2 + 2 \cdot \text{VaR}_1 \cdot \text{VaR}_2 \cdot w_1 \cdot w_2 \cdot \rho}$$

Substituting the given values:

$$\text{VaR}_p = \sqrt{\left(\frac{1}{2}\right)^2 \cdot 10^2 + \left(\frac{1}{2}\right)^2 \cdot 10^2 + 2 \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 10 \cdot 10 \cdot \rho}$$

This simplifies to:

$$\text{VaR}_p = \sqrt{\frac{1}{4} \cdot 100 + \frac{1}{4} \cdot 100 + 2 \cdot \frac{1}{4} \cdot 100 \cdot \rho}$$

$$\text{VaR}_p = \sqrt{50 + 50 + 50 \cdot \rho}$$

Thus, the portfolio VaR in terms of the correlation ρ is:

$$\text{VaR}_p = \sqrt{100 + 50 \cdot \rho}$$

If $\rho = 0$, this simplifies to $\text{VaR}_p = 10$, as shown in the original example.

7.3 Expected Shortfall

Introduction

Expected Shortfall (ES), also known as Conditional Value at Risk (CVaR), is a risk measure used in finance to quantify the expected loss in extreme market conditions, beyond a certain Value at Risk (VaR) threshold. ES is commonly used to better understand tail risks, particularly in cases of market crashes or financial crises.

Formula

The mathematical definition of Expected Shortfall (ES) at a confidence level α is given as:

$$ES_{\alpha} = -E[L|L > VaR_{\alpha}]$$

Where:

- L is the loss variable,
- VaR_{α} is the Value at Risk at confidence level α ,
- $E[L|L > VaR_{\alpha}]$ is the conditional expectation of the loss L given that the loss exceeds the VaR_{α} threshold.

Steps for Calculation

To compute the Expected Shortfall, follow these steps:

1. **Select a confidence level α :** Common choices for α include 95% or 99%.
2. **Calculate Value at Risk (VaR):** Find the maximum loss at the given confidence level.
3. **Calculate the conditional expectation of losses beyond VaR:** Average the losses that exceed the VaR threshold. This conditional mean represents the Expected Shortfall.

Example

Suppose we have daily loss data and we wish to calculate the Expected Shortfall at the 99% confidence level:

- **Confidence Level:** $\alpha = 99\%$.
- **VaR Calculation:** Assume the $VaR_{99} = 1000$ USD.
- **Expected Shortfall Calculation:** Find the average of all losses greater than 1000 USD. Assume these losses average to 1200 USD.

Thus, the Expected Shortfall at the 99% confidence level is 1200 USD.

7.4 VaR Calculation

What is the ten-day 99% VaR of a portfolio with a five-day 98% VaR of \$10 million?

Answer:

If we assume normally distributed short term portfolio returns, the VaR (Value at Risk) of a portfolio is proportional to the square root of the time horizon. More precisely, if the N -day $C\%$ VaR of the portfolio is denoted by $VaR(N, C)$, where N is the number of days in the time horizon and C is the confidence level, then

$$VaR(N, C) \approx \sigma_V z_C \sqrt{\frac{N}{252}} V(0),$$

where σ_V is the (annualized) standard deviation of the rate of return of the portfolio, z_C is the z -score of the standard normal distribution corresponding to C , i.e., $P(Z \leq z_C) = C$, and $V(0)$ is the current value of the portfolio.

From (3.76), it follows that

$$\text{VaR}(10 \text{ days}, 99\%) \approx \frac{z_{99}\sqrt{10}}{z_{98}\sqrt{5}} \text{VaR}(5 \text{ days}, 98\%).$$

For $\text{VaR}(5 \text{ days}, 98\%) = \$10,000,000$, and since $z_{99} \approx 2.326348$ and $z_{98} \approx 2.053749$, we obtain that

$$\text{VaR}(10 \text{ days}, 99\%) \approx \$16,019,255. \quad \square$$